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(54) **INTERCONNECT STRUCTURE WITHOUT BARRIER LAYER ON BOTTOM SURFACE OF VIA**

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(71) Applicant: **Taiwan Semiconductor Manufacturing Co., Ltd.**, Hsinchu City (TW)

(72) Inventors: **Tz-Jun KUO**, Hsinchu City (TW);
Chien-Hsin Ho, Taichung City (TW);
Ming-Han Lee, Taipei City (TW)

(73) Assignee: **Taiwan Semiconductor Manufacturing Co., Ltd.**, Hsinchu City (TW)

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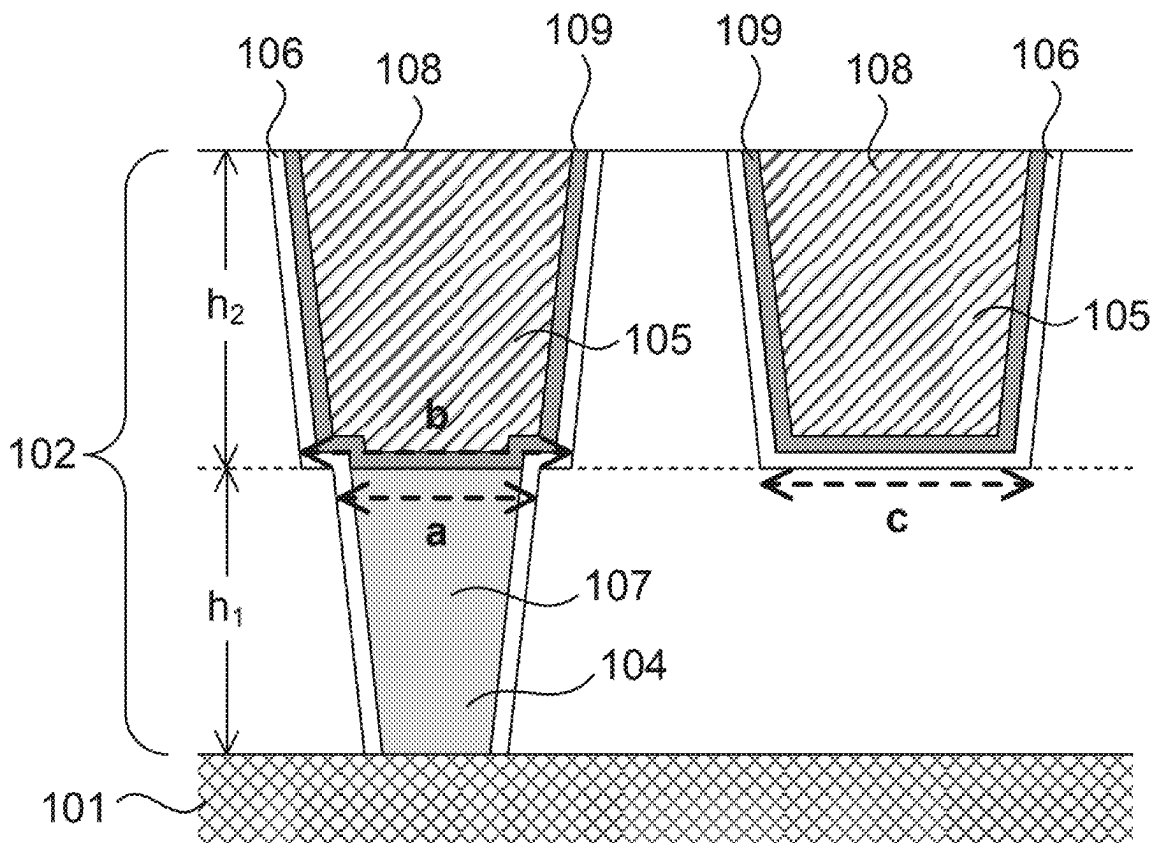
(22) Filed: **May 8, 2025**

Related U.S. Application Data

- (60) Continuation of application No. 17/734,683, filed on May 2, 2022, now Pat. No. 12,308,282, which is a division of application No. 16/593,562, filed on Oct. 4, 2019, now Pat. No. 11,322,391, which is a division of application No. 15/679,385, filed on Aug. 17, 2017, now Pat. No. 10,453,740.
- (60) Provisional application No. 62/526,420, filed on Jun. 29, 2017.

(57) **ABSTRACT**

Embodiments and methods of an interconnect structure are provided. The interconnect structure includes a via, a trench that has an overlapping area with a top of the via, and a first layer of conducting material that has an overlapping area with a bottom of the via. The interconnect also includes a second layer of conducting material formed in the via, and a third layer of conducting material formed in the trench. The second layer of conducting material is in contact with the first layer of conducting material without a barrier in between the two conducting materials. The absence of the barrier at the bottom of the via can reduce the contact resistance of the interconnect structure.



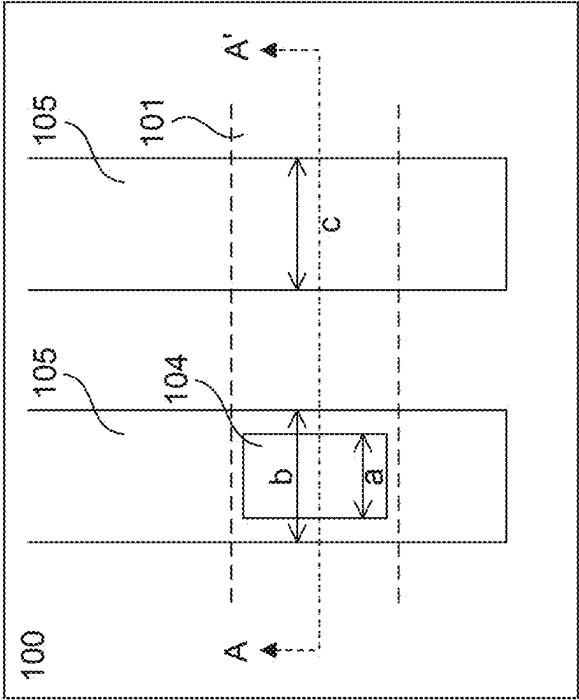


FIG. 1A

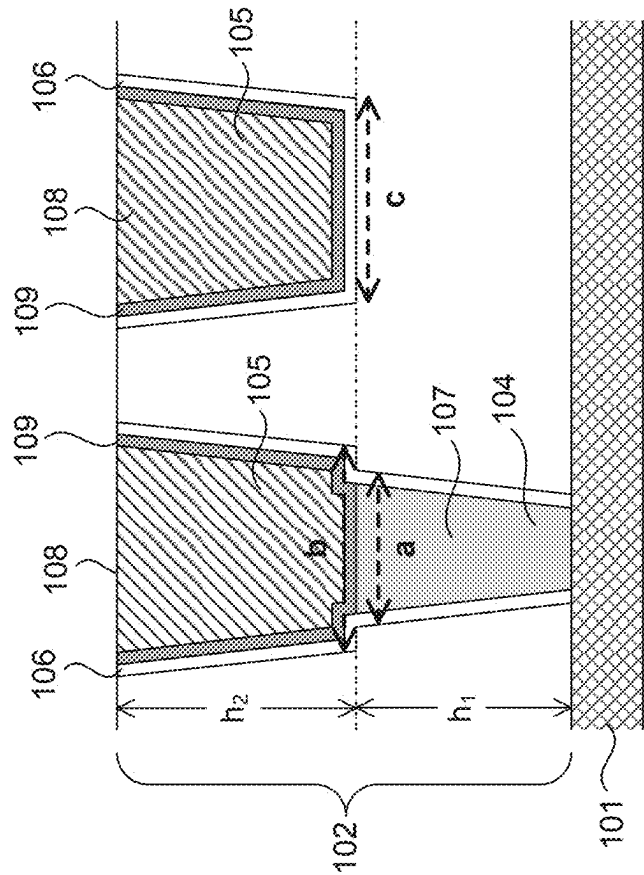


FIG. 1B

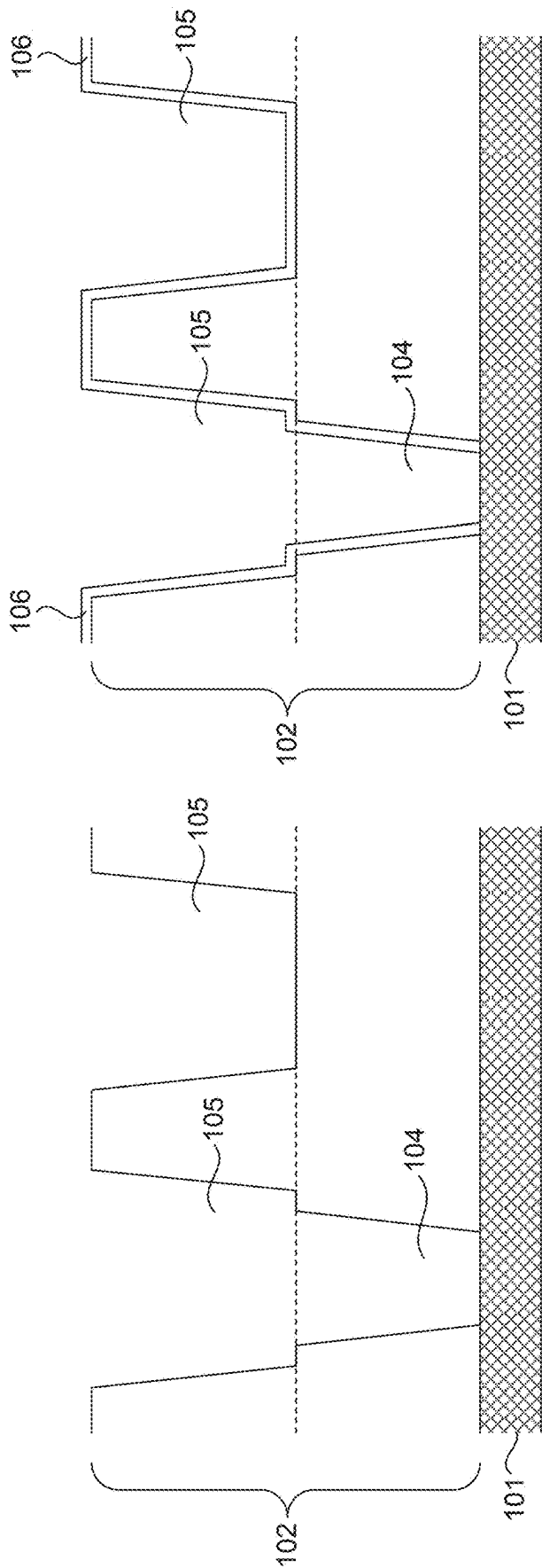


FIG. 2B

FIG. 2A

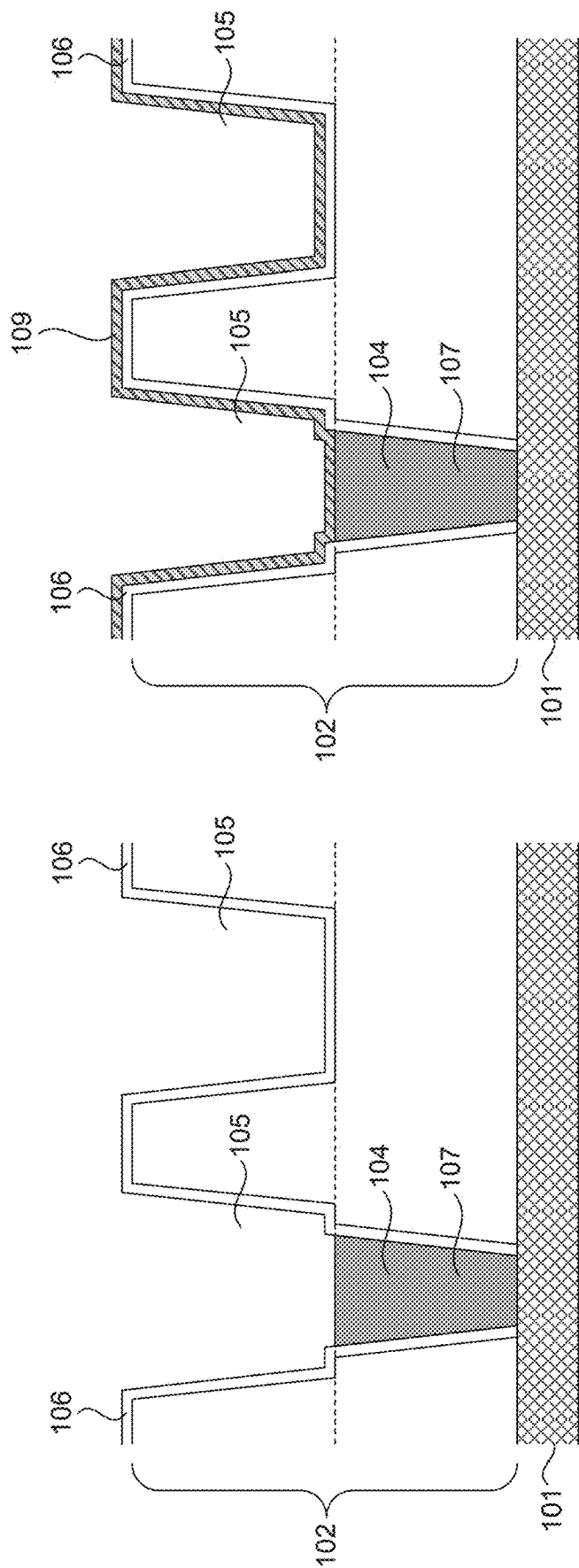


FIG. 2D

FIG. 2C

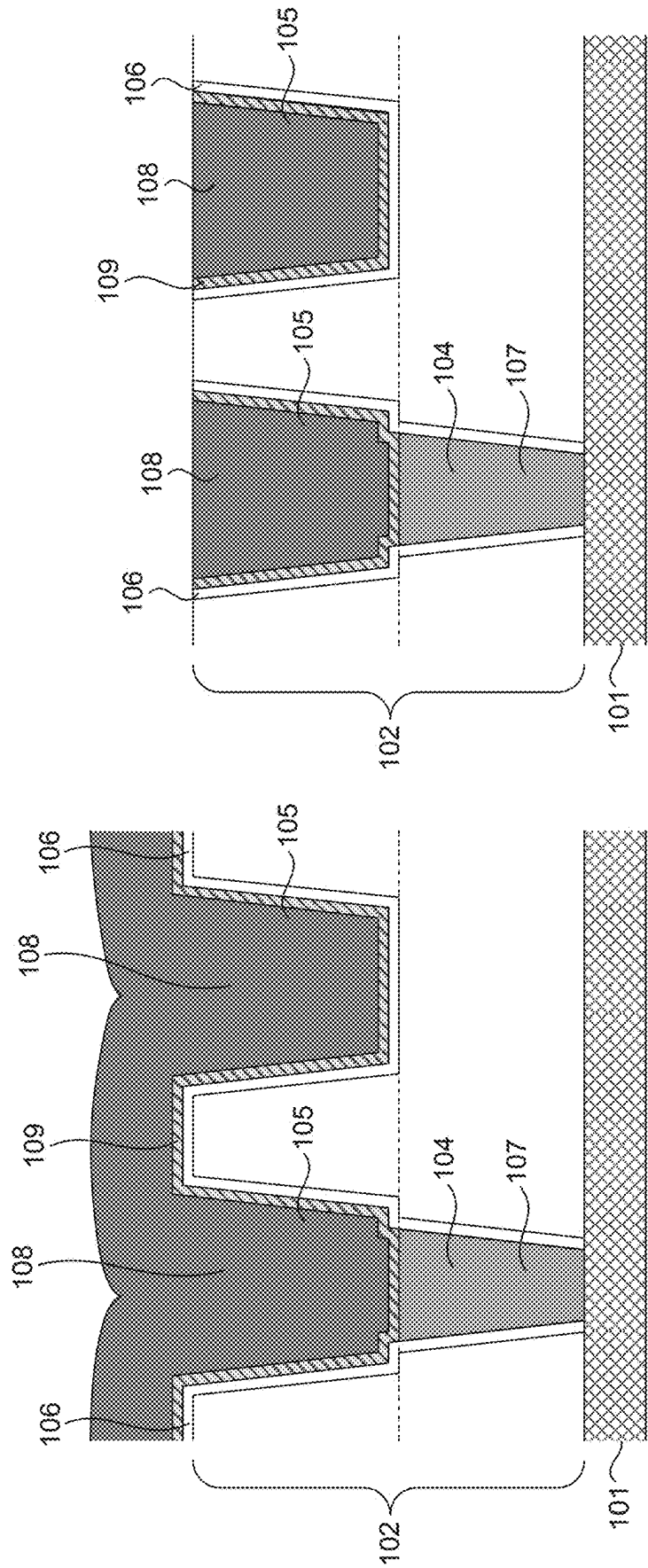


FIG. 2E

FIG. 2F

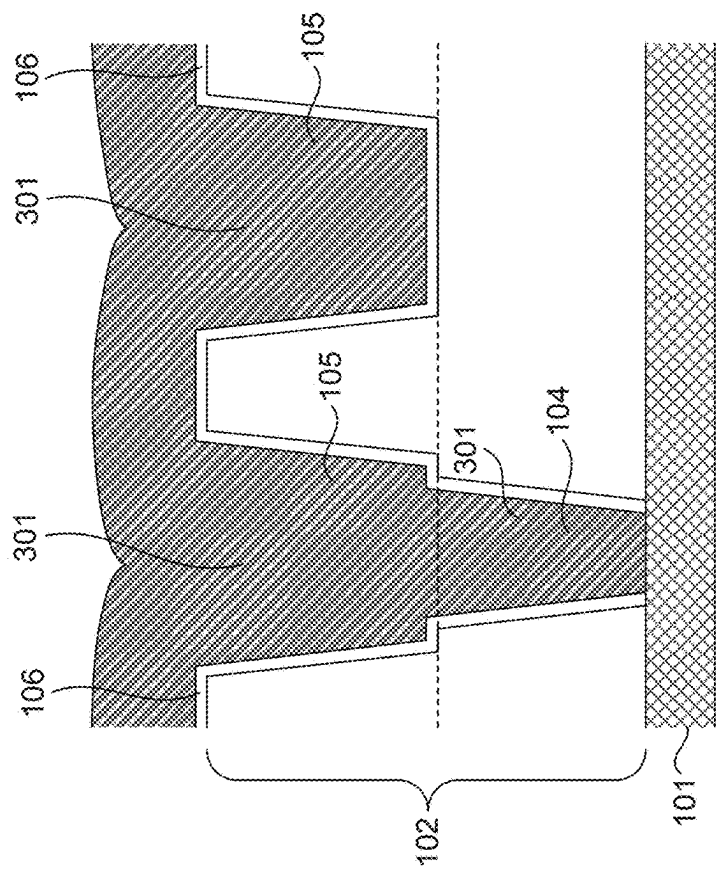


FIG. 3A

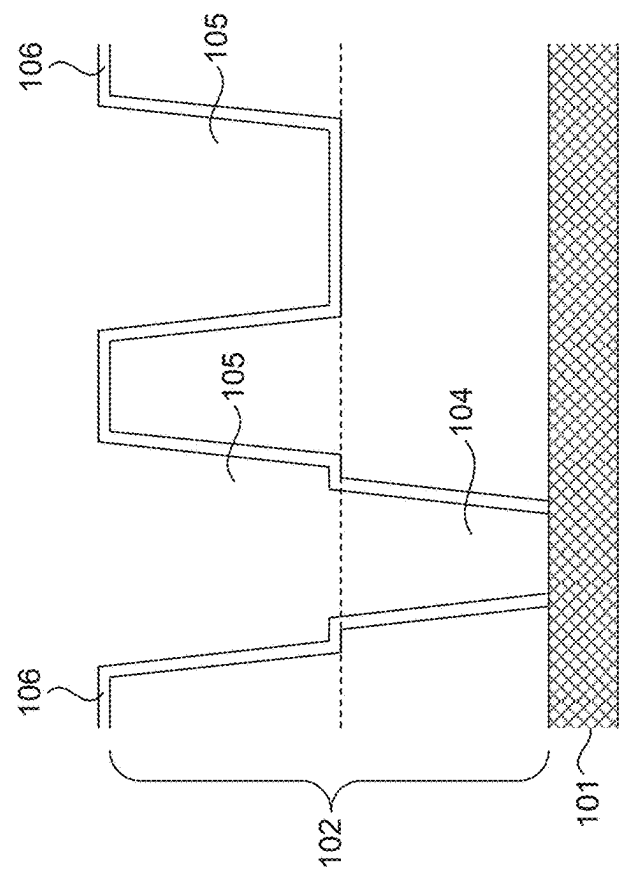


FIG. 3B

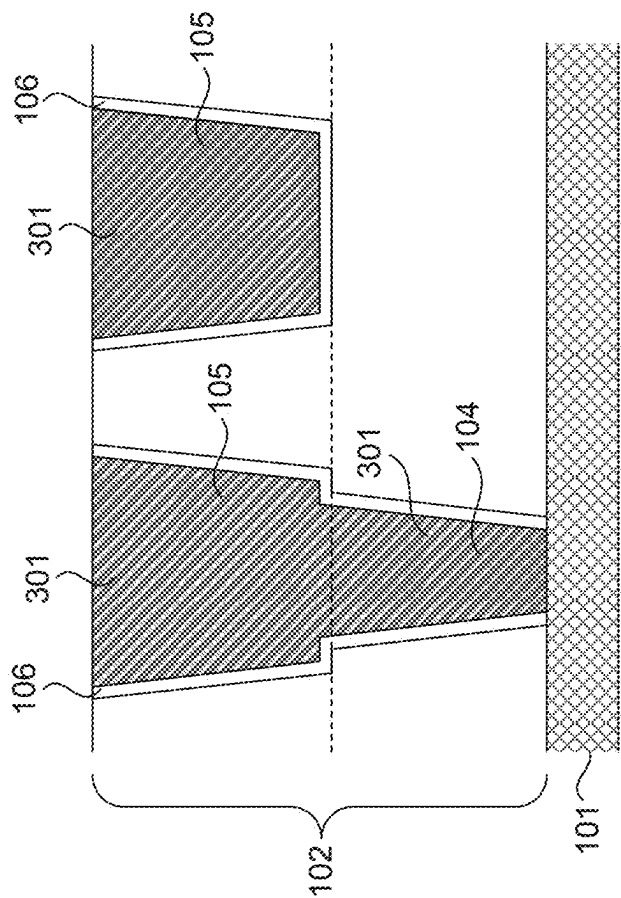
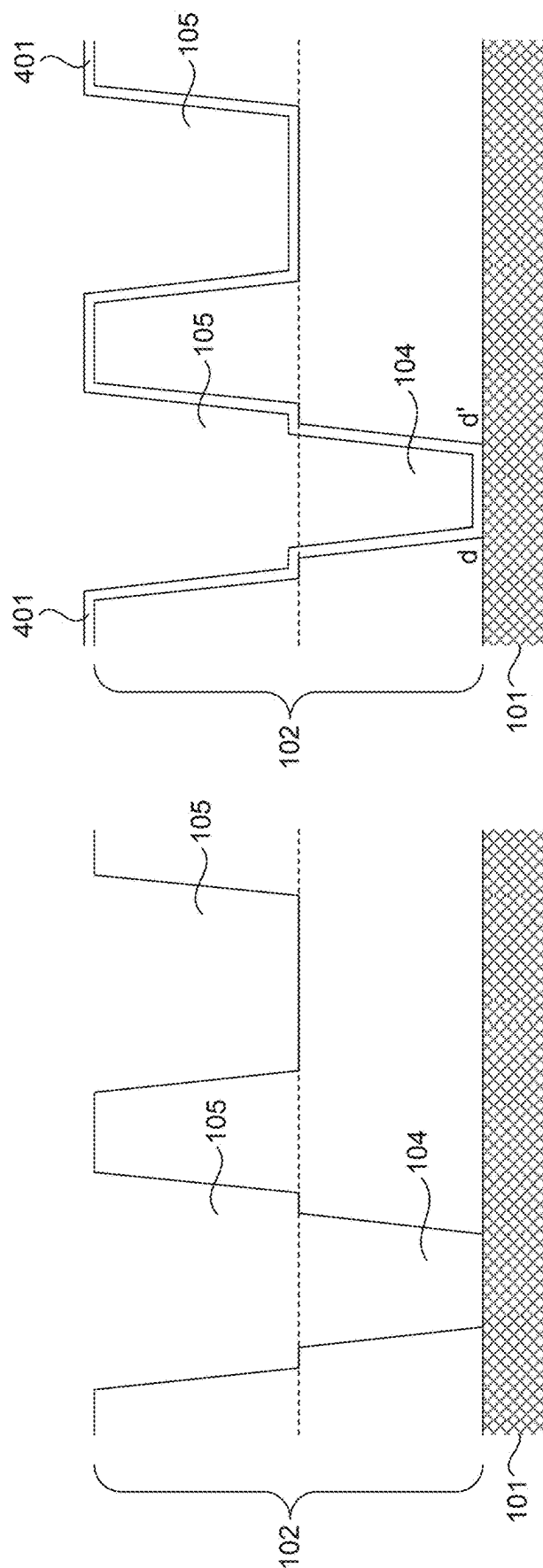
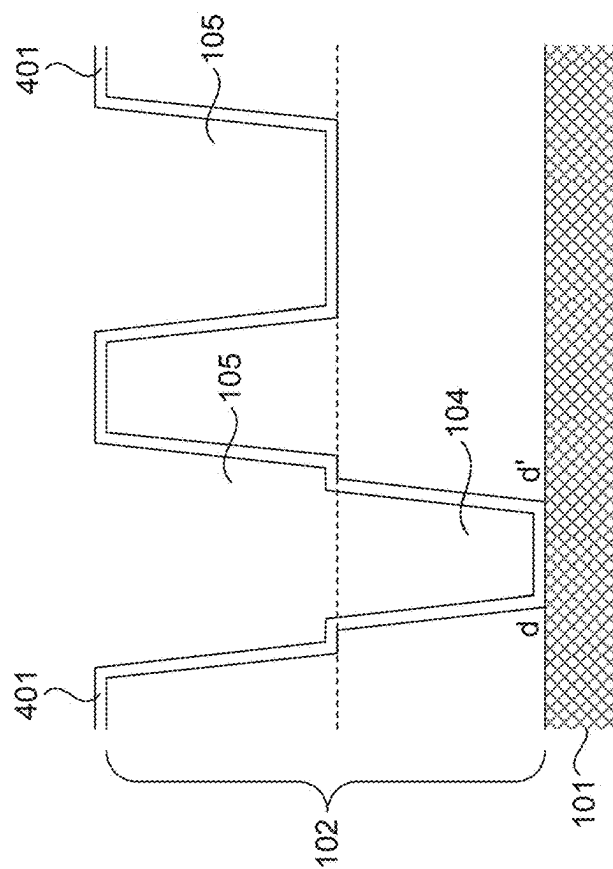


FIG. 3C



4A
G
E



മുഖ്യമന്ത്രി

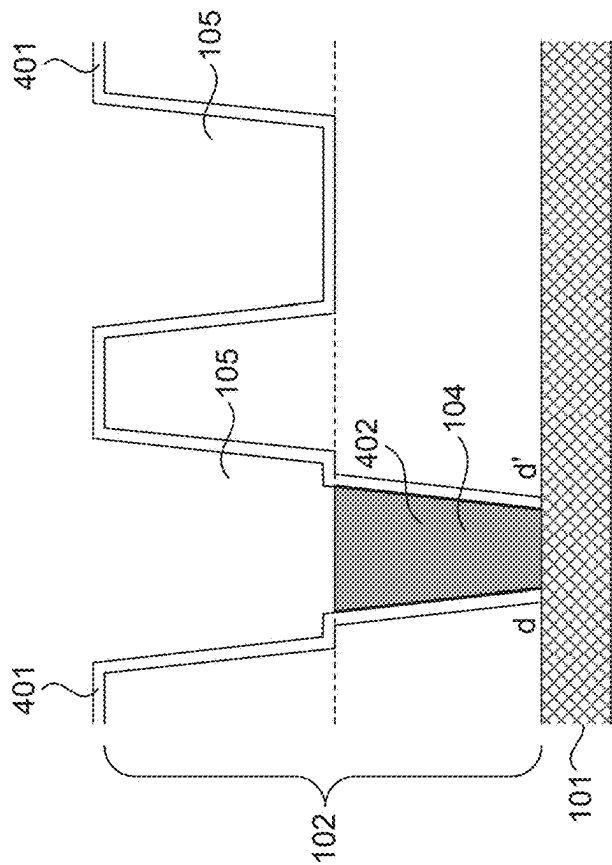


FIG. 4C

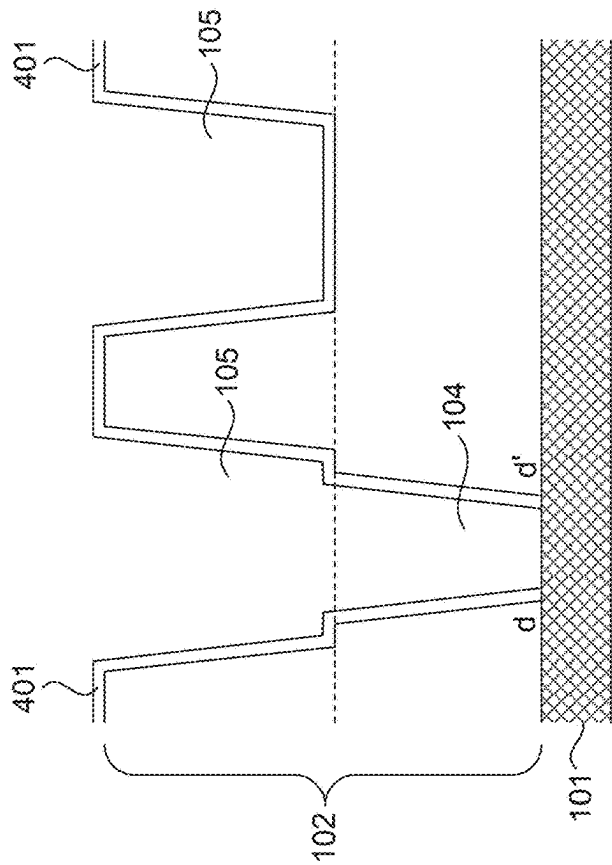


FIG. 4D

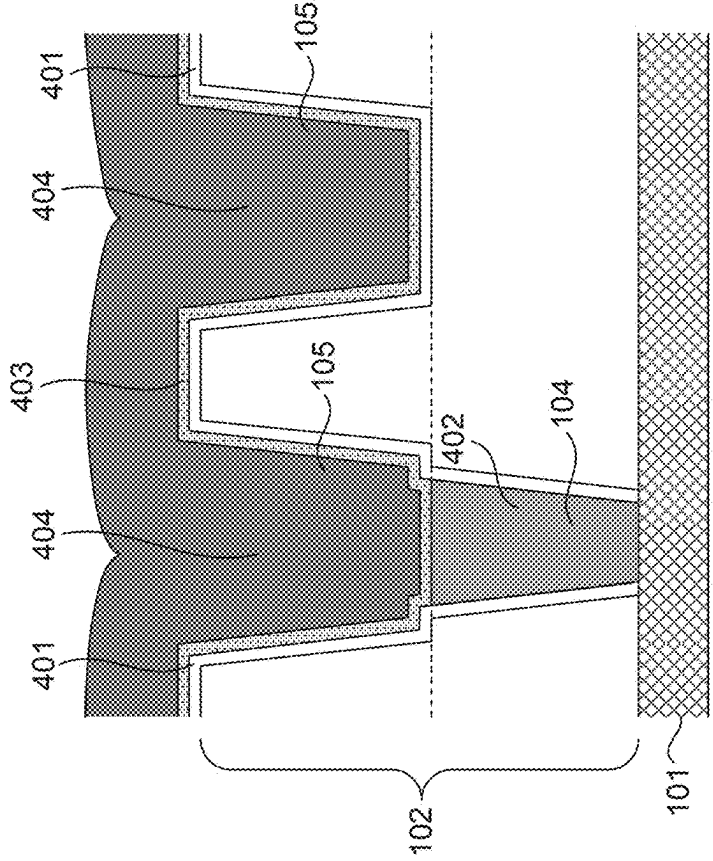


FIG. 4E

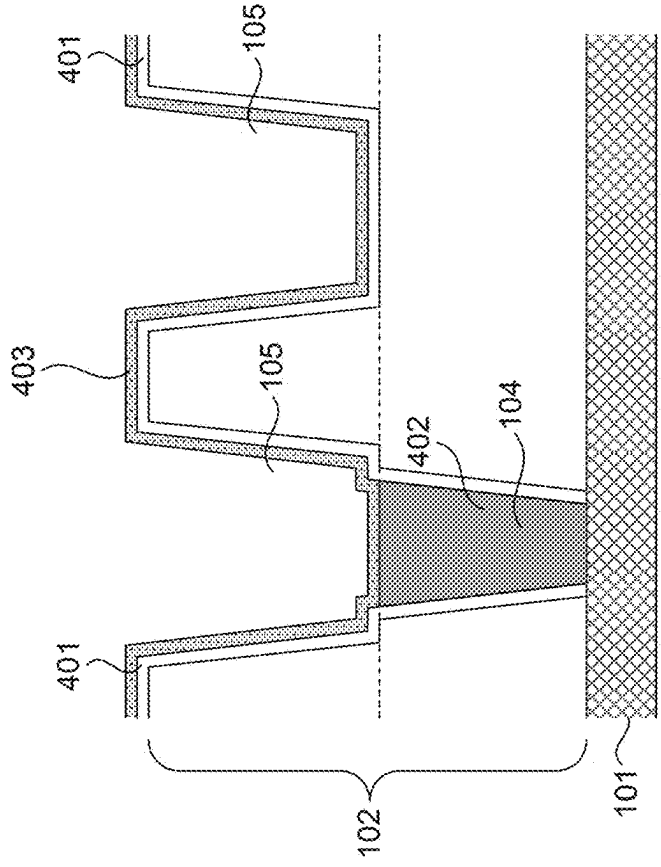


FIG. 4F

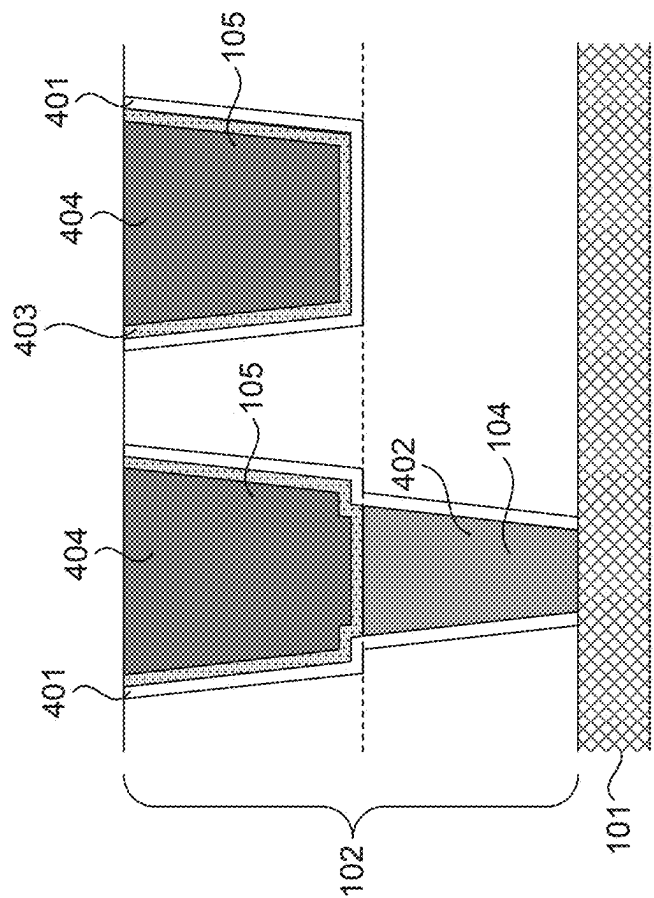


FIG. 4G

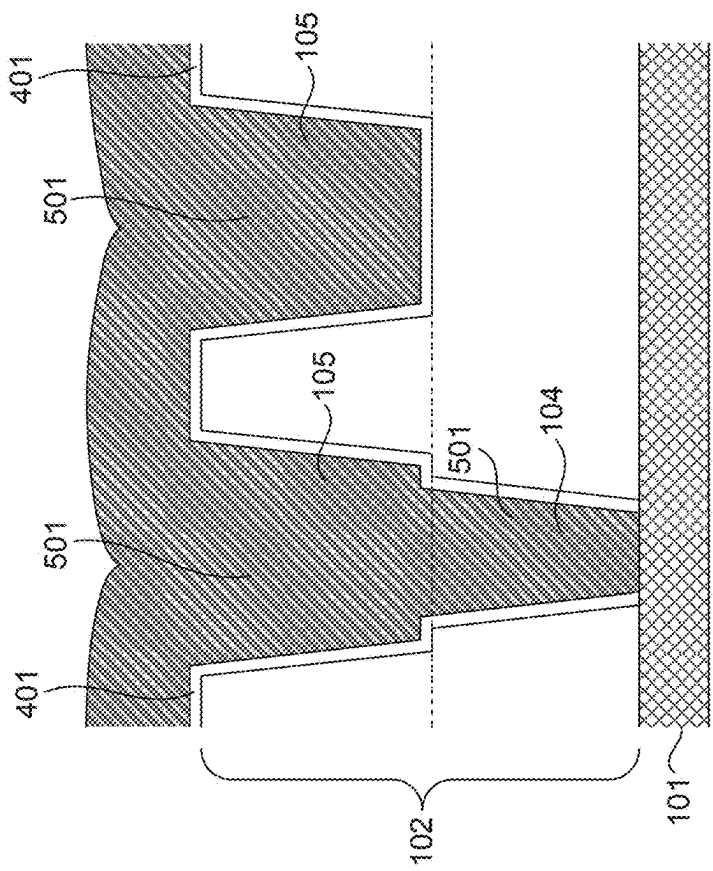


FIG. 5A

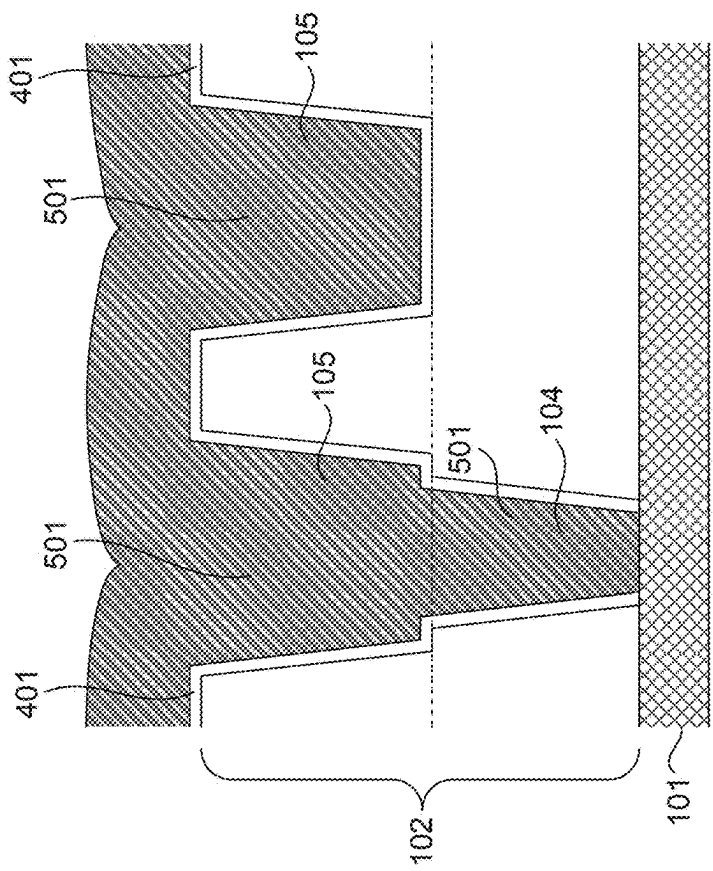


FIG. 5B

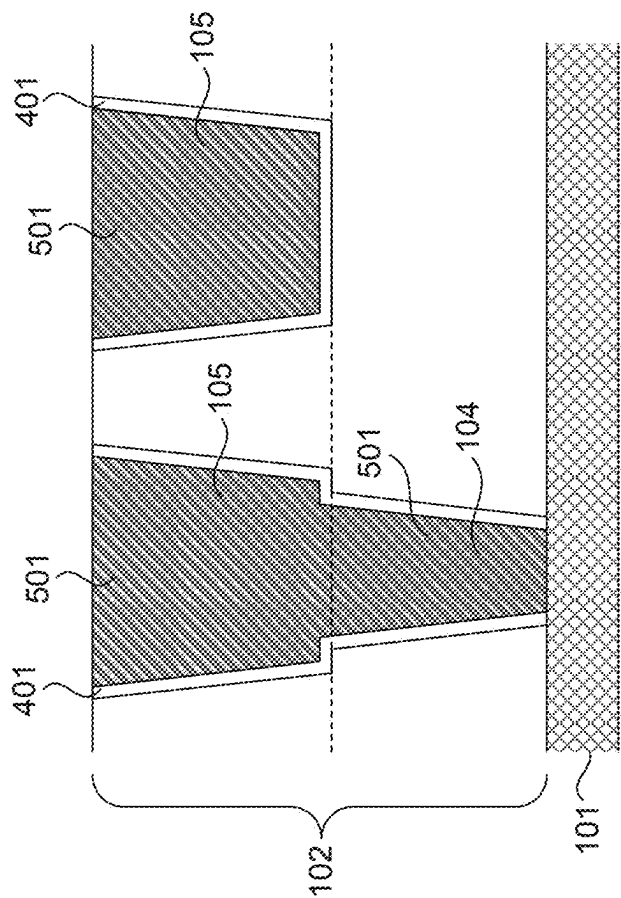


FIG. 5C

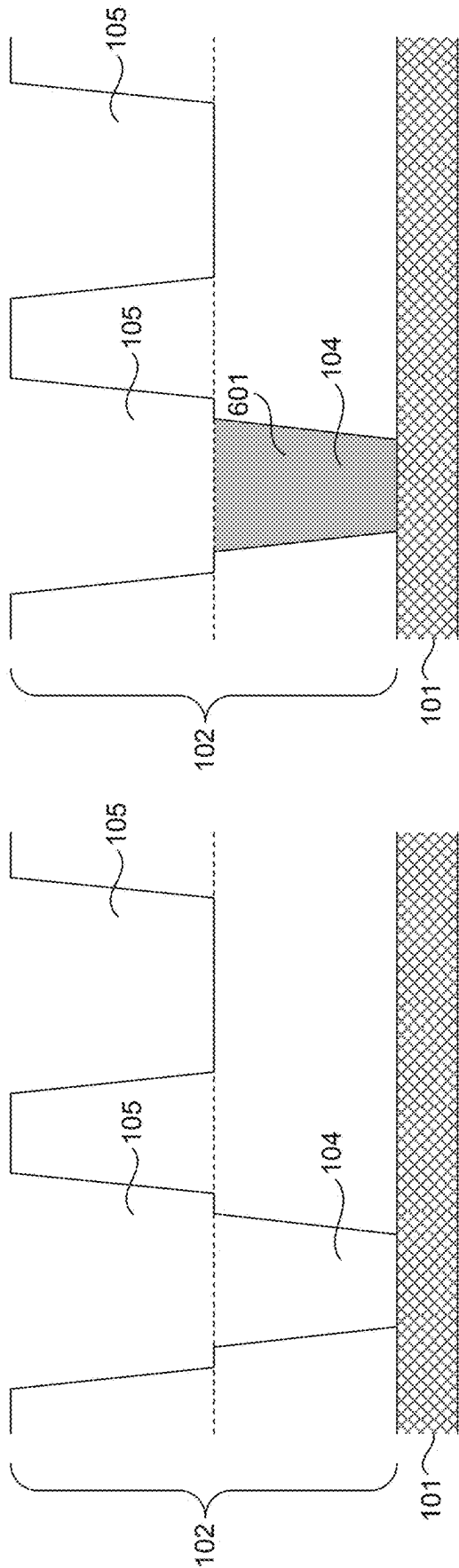


FIG. 6A

FIG. 6B

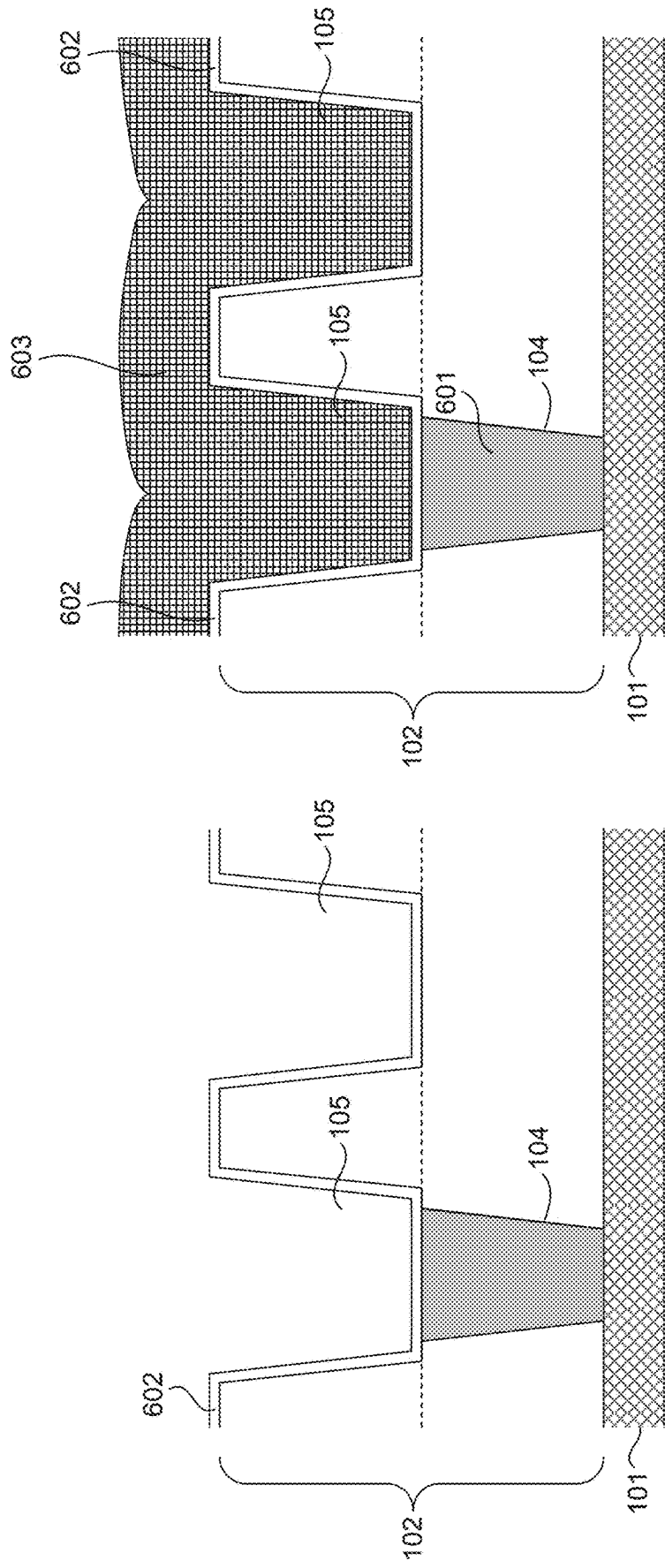


FIG. 6C

FIG. 6D

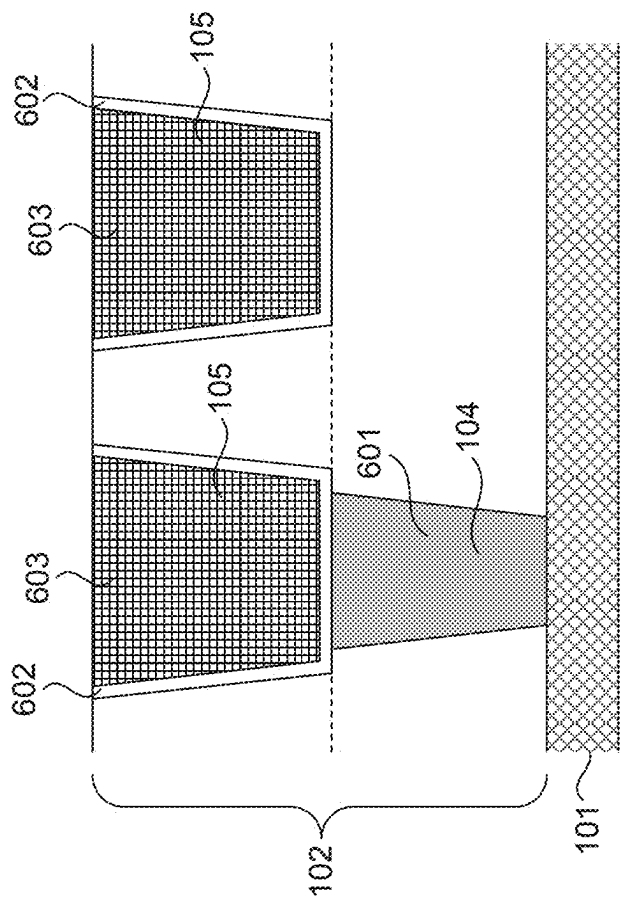


FIG. 6E

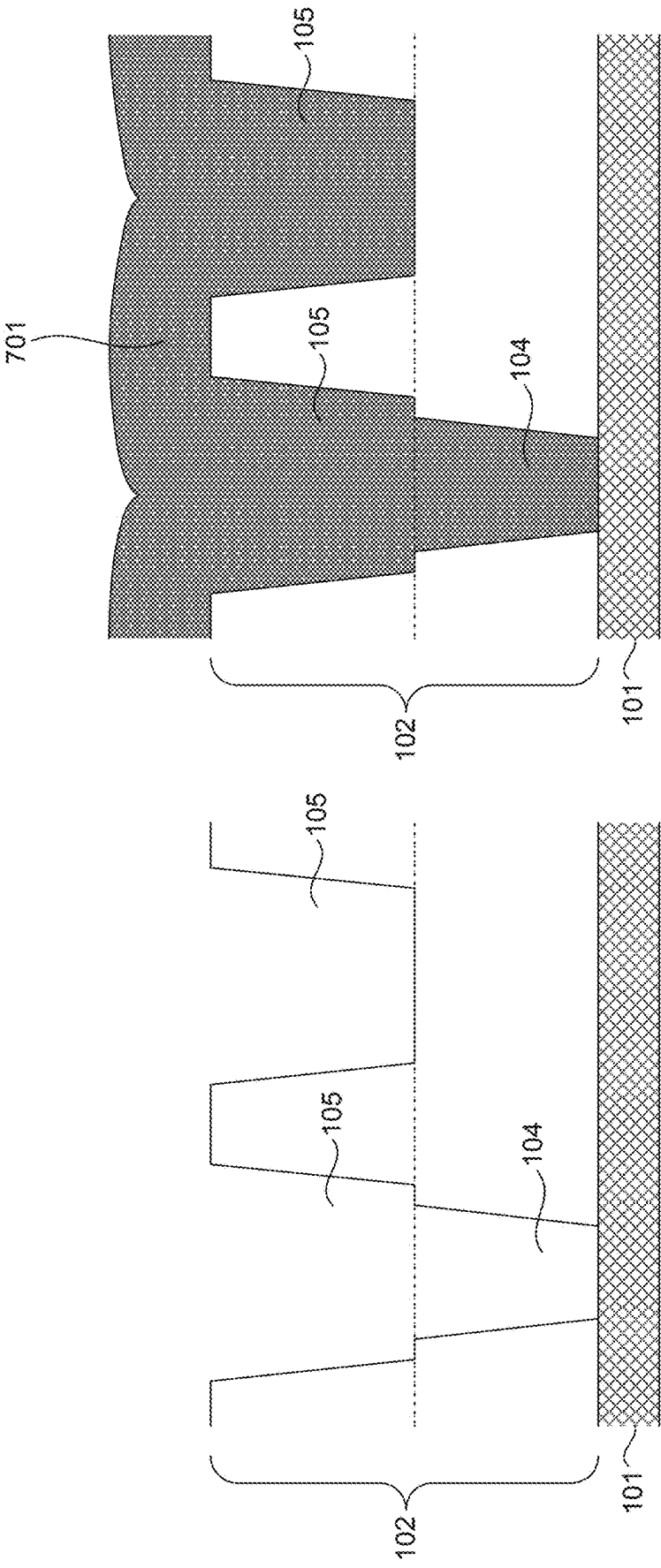


FIG. 7A

FIG. 7B

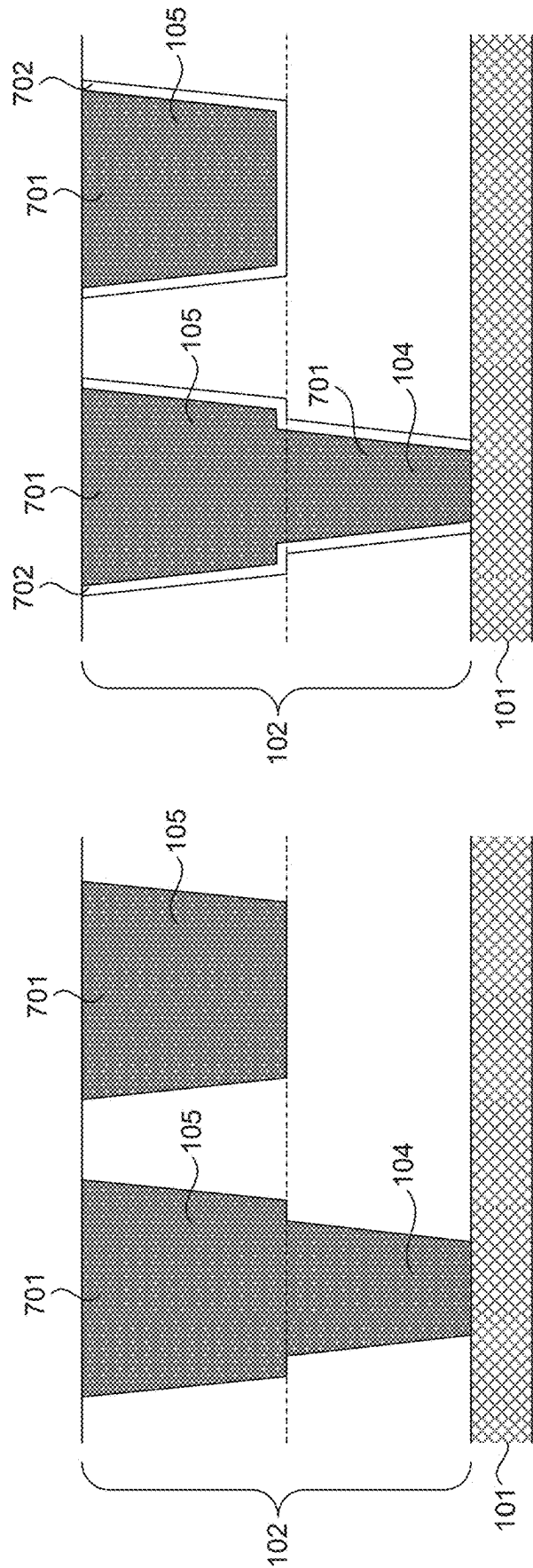


FIG. 7D

FIG. 7C

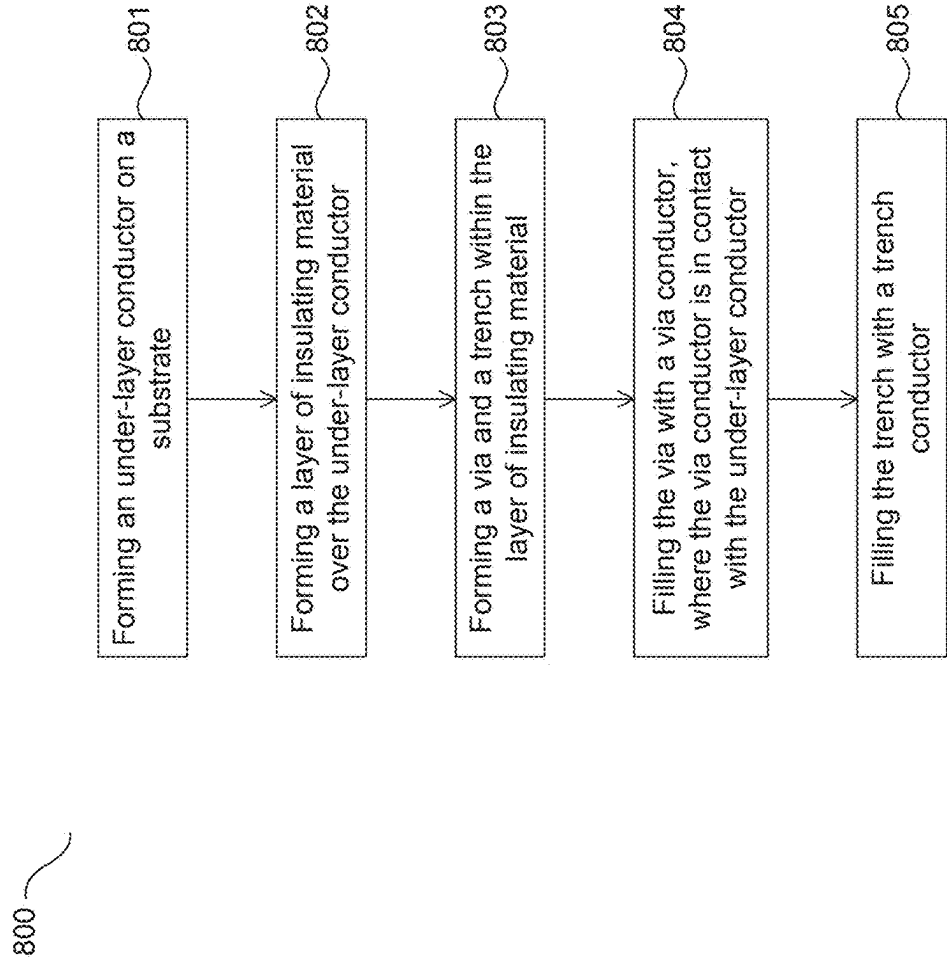


FIG. 8

INTERCONNECT STRUCTURE WITHOUT BARRIER LAYER ON BOTTOM SURFACE OF VIA

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. Non-provisional patent application Ser. No. 17/734,683, filed on May 2, 2022, titled “Interconnect Structure Without Barrier Layer on Bottom Surface of Via,” which is a divisional of U.S. Non-Provisional patent application Ser. No. 16/593,562, filed on Oct. 4, 2019, titled “Interconnect Structure Without Barrier Layer on Bottom Surface of Via,” which is a divisional of U.S. Non-Provisional patent application Ser. No. 15/679,385, filed on Aug. 17, 2017, titled “Interconnect Structure Without Barrier Layer on Bottom Surface of Via,” which claims the benefit of U.S. Provisional Patent Application No. 62/526,420, filed on Jun. 29, 2017, titled “Interconnect Structure Without Barrier Layer on Bottom Surface of Via.” The entire contents of the aforementioned applications are incorporated by reference herein in their entireties.

BACKGROUND

[0002] With advances in semiconductor technology, interconnect dimensions are shrinking to accommodate increasing device densities. Such scaling down in interconnect dimensions has increased the complexity of semiconductor manufacturing processes for fabricating interconnects with low resistance and high reliability.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of this disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the common practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1A is a top down view of an exemplary interconnect structure according to some embodiments.

[0005] FIG. 1B is a cross-sectional view of FIG. 1A along cross sectional line A-A', according to some embodiments.

[0006] FIGS. 2A to 2F are cross sectional views of exemplary interconnect structures during formation, according to some embodiments.

[0007] FIGS. 3A to 3C are cross sectional views of an exemplary interconnect structure formed with a dual damascene process, according to some embodiments.

[0008] FIGS. 4A to 4G are cross sectional views of exemplary interconnect structures with a selective removal of a portion of a via-plus-trench barrier, according to some embodiments.

[0009] FIGS. 5A to 5C are cross sectional views of an exemplary interconnect structure formed with a dual damascene process, according to some embodiments.

[0010] FIGS. 6A to 6E are cross sectional views of forming exemplary interconnect structures without forming a selective via-plus-trench barrier, according to some embodiments.

[0011] FIGS. 7A to 7D are cross sectional views of forming exemplary interconnect structures with a self-forming barrier, according to some embodiments.

[0012] FIG. 8 is a flow diagram of an example method of forming exemplary interconnect structures, according to some embodiments.

[0013] Illustrative embodiments will now be described with reference to the accompanying drawings. In the drawings, like reference numerals generally indicate identical, functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

[0014] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to exemplify the present disclosure. These examples are not intended to be limiting. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0015] Spatially relative terms, Such as “on,” “in,” “over,” “beneath,” “below,” “lower,” “above,” “upper,” “top,” “bottom,” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0016] It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “exemplary,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

[0017] It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.

[0018] The term “about” as used herein indicates the value of a given quantity varies by $\pm 10\%$ of the value, unless noted otherwise.

[0019] As used herein, the term “selectivity” refers to the ratio of growth rates, deposition rates, etching rates or removal rates between two materials under the same process conditions. The term “selective,” “selectively,” “exclusive,” and “exclusively” are used when the growth rate, deposition rate, etching rate or removal rate of a material on an intended surface is at least 10 times higher than that on an un-intended surface. The term “minimal” is used when the thickness of a material is less than the thickness of a monolayer of the material.

[0020] As used herein, the term “substrate” describes a material onto which subsequent material layers are added. The substrate itself may be patterned, and materials added

on top of it may also be patterned, or may remain without patterning. Furthermore, “substrate” may be any of a wide array of semiconductor materials such as, for example, silicon, germanium, gallium arsenide, indium phosphide, etc. Alternatively, the substrate may be an electrically non-conductive material such as, for example, glass or sapphire wafer.

[0021] As used herein, the term “high-k” refers to a high dielectric constant. In the field of semiconductor device structures and manufacturing processes, high-k refers to a dielectric constant that is greater than the dielectric constant of SiO₂ (i.e., greater than 3.9).

[0022] As used herein, the term “vertical” means nominally perpendicular to the surface of a substrate.

[0023] As used herein, the term “via” refers to a vertical hole that has a top, a bottom and a sidewall. The top of a via refers to a higher boundary of the via, and the bottom of a via refers to a lower boundary of the via. Lower and higher both refer to vertical locations, with the substrate at lower direction.

[0024] As used herein, the term “trench” refers to a box like structure with a top, a bottom, and at least a sidewall. A trench usually has a length that is larger than its width. A trench does not necessarily have sharp corners or sharp edges. A trench can have different sections with different sizes, different shapes, and different length directions.

[0025] As used herein, the term “form” or “formed” refers to one or more steps of adding or removing one or more components.

[0026] This disclosure provides various exemplary interconnect structures in semiconductor devices. This disclosure also provides exemplary methods for fabricating the interconnect structures with lower interconnect resistance and lower contact resistance in small dimension and high aspect ratio structures.

[0027] FIG. 1A is a top down view of an exemplary interconnect structure, according to some embodiments. FIG. 1B is a cross-sectional view of FIG. 1A along cross sectional line A-A', according to some embodiments. In referring to FIG. 1A, an interconnect structure is formed above a substrate **100**. In some embodiments, an under-layer conductor **101** can be formed above the substrate **100**. For example, under-layer conductor **101** can be formed to be in contact with the substrate **100**. Alternatively, one or more intermediate layers of, for example, an interconnect or an insulating material can be formed between the under-layer conductor **101** and the substrate **100**. In some embodiments, the under-layer conductor **101** can be a patterned layer of metal for an interconnect. This patterned layer of metal can include metals and/or barriers to prevent diffusion of the metal into the under-layer conductor **101**. In some embodiments, the under-layer conductor **101** can be a patterned layer of conducting area formed on silicon (Si) such as, for example, implanted source or drain areas for a field-effect-transistor (FET) device.

[0028] In referring to FIG. 1B, a layer of insulating material **102** is formed above the under-layer conductor **101**. The insulating material **102** can be made from insulating materials. The insulating materials include, for example, non-doped silicon dioxide, doped silicon dioxide, silicon nitride with or without dopants, and silicon oxy-nitride with and without dopants. In some embodiments, the insulating

material **102** can include a plurality of layers made from different insulating materials, according to some embodiments.

[0029] In referring to FIG. 1A, a via **104** and a trench **105** can be formed in the layer of insulating material **102**. In referring to FIG. 1B, the cross-sectional view illustrates that the bottom of the via **104** contacts a portion of the top surface of the under-layer conductor **101**. The top of the via **104** contacts a portion of a bottom of the trench **105**. FIGS. 1A and 1B show that the width of the bottom of the trench **105** (annotated as ‘b’ in FIG. 1A) is larger than the width of the top of the via **104** (annotated as ‘a’ in FIG. 1A) at an overlapping area of the two interconnect structures. The width of the trench **105** can be larger than, smaller than, or the same as the width of the via **104**. The width of the trench **105** (annotated as ‘c’ in FIG. 1A) in a non-overlapping area can also be larger than, smaller than, or the same as the width b of the trench **105** at the overlapping area. In some embodiments, each of the widths a, b, and c is between 1 nm and 50 nm. For example, each of the widths a, b, and c can be between 10 and 30 nm (e.g., about 20 nm). In some embodiments, the average widths of trench can be between 10 and 30 nm. In some embodiments, the aspect ratio (i.e., the ratio between the height and the width) of the via can be between 0.5 and 100. For example, the aspect ratio of the via can be greater than 10. In some embodiments, the aspect ratio of the trench can be between 0.5 and 10. In some embodiments, via **104** and trench **105** can have respective heights h₁ and h₂. In some embodiments, the height ratio h₁/h₂ can be less than about 1 or greater than about 20. In some embodiments, the height ratio h₁/h₂ can be in a range of about 1 and about 20. In referring to FIG. 1A, a top portion of the via has a rectangular shape. The via can take different shapes (e.g., circular, oval-shaped, rectangular with rounded corners, etc.).

[0030] Referring to FIG. 1B, in some embodiments, a layer of via-plus-trench barrier **106** is formed after forming the via **104** and the trench **105**. In some embodiments, the layer of via-plus-trench barrier **106** is formed on the sidewalls of the trench **105**, the bottom of the trench **105**, and the sidewalls of the via **104**. Minimal or no via-plus-trench barrier **106** is formed over the overlapping area between the via **104** and the under-layer conductor **101**.

[0031] In some embodiments, the thickness of the via-plus-trench barrier **106** thickness is between 1 Å and 50 Å. In some embodiments, the thickness of the selective via-plus-trench barrier **106** thickness is between 5 Å and 30 Å or between 15 Å and 25 Å. As interconnect dimensions are shrinking to accommodate increasing device densities, a via-plus-trench barrier layer with lower thickness can be selected to increase the proportion of conductive material in the interconnect structures. For example, the thickness of via-plus-trench barrier **106** can be between 1 Å and 20 Å. The via-plus-trench barrier **106** can include a metal (e.g., tantalum (Ta), titanium tungsten (TiW), and/or other metal or metal alloys), a metal oxide (e.g., alumina, manganese oxide, chromium oxide, niobium oxide, titanium oxide, and/or combinations thereof), a metal nitride (e.g. TaN, titanium nitride (TiN)), an organo-silane (e.g., trimethoxysilane with long alkyl chain), an organo-phosphane, other suitable materials, and/or combinations thereof.

[0032] Referring to FIG. 1B, in some embodiments, a trench barrier **109** is formed on the sidewalls of the trench **105** and the bottom of the trench **105**. The trench barrier **109**

is in contact with the via-plus-trench barrier **106** and the top surface of a via conductor **107**.

[0033] In some embodiments, the trench barrier **109** includes a metal (e.g., tantalum (Ta)), metal (e.g., Ta, TiW, and/or other metal or metal alloys), a metal oxide (e.g., alumina, manganese oxide, chromium oxide, niobium oxide, titanium oxide, and/or combinations thereof), a metal nitride (e.g., TaN, titanium nitride), a metal compound (e.g., alumina, manganese oxide, chromium oxide, niobium oxide, titanium oxide, and/or combinations thereof), a carbon containing material (e.g., graphene), other suitable materials, and/or combinations thereof. For example, a graphene-based trench barrier layer **109** can be formed through any suitable methods such as, for example, by chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), other deposition processes, and/or combinations thereof. In some embodiments, a passivation layer is deposited on exposed metal surfaces prior to the formation of a graphene-based trench barrier layer. The passivation layer can be formed of any suitable material such as, for example, aromatic compounds, long carbon chain with functional groups, other suitable materials, and/or combinations thereof. In some embodiments, the functional groups can include acid, amine, phosphine, other suitable functional groups, and/or combinations thereof. In some embodiments, a catalyst layer can also be formed on exposed dielectric surfaces in trench **105** prior to the deposition of graphene material. In some embodiments, the catalyst layer is not formed on the surfaces of via conductor **107** in trench **105**. The catalyst layer can be formed of iron (Fe), nickel (Ni), cobalt (Co), other suitable material, and/or combinations thereof. The passivation layer and the catalyst layer can react with precursors in a CVD or PECVD process to form a graphene-based trench barrier layer. In some embodiments, the thickness of the trench barrier **109** is between 1 Å and 50 Å. In some embodiments, the thickness of the trench barrier **109** is between 5 Å and 30 Å. In some embodiments, the thickness of the trench barrier **109** is between 15 Å and 25 Å. In some embodiments, the thickness of trench barrier **109** is between 1 Å and 20 Å.

[0034] In referring to FIG. 1B, the via **104** is filled with a via conductor **107**. The trench **105** is filled with a trench conductor **108**. As illustrated by FIG. 1B, the via conductor **107** is in contact with the under-layer conductor **101** and the via-plus-trench barrier **106**. The trench conductor **108** is in contact with the trench barrier **109**. The trench conductor **108** is electrically coupled to the via conductor **107** at the overlapping area between the trench **105** and the via **104**.

[0035] As illustrated in FIG. 1B, a top surface of the via conductor **107** is at or approximately at same level as a top interface of the via **104**. In some embodiments, the top surface of the via conductor **107** can be higher or lower than the top interface of the via **104**.

[0036] FIGS. 2A to 2F are cross sectional views of exemplary interconnect structures during formation of the interconnect structure shown in FIG. 1B, according to some embodiments. In some embodiments, the interconnect structure is formed on the substrate **100** (not shown in FIGS. 2A to 2F). The under-layer conductor **101** is formed on the substrate **100**. In some embodiments, the under-layer conductor **101** includes copper (Cu), cobalt (Co), nickel (Ni), ruthenium (Ru), rhodium (Rh), iridium (Ir), osmium (Os), aluminum (Al), indium (In), tungsten (W), or carbon nanotubes. In some embodiments, the under-layer conductor **101**

includes metals alloyed with boron (B), aluminum (Al), titanium (Ti), chromium (Cr), manganese (Mn), niobium (Nb), palladium (Pd), silver (Ag), indium (In), or gold (Au). The layer of insulating material **102** is formed over the under-layer conductor **101**. In some embodiments, the layer of insulating material **102** is formed by chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), a spin-on process, physical vapor deposition (PVD), or atomic layer deposition (ALD). In some embodiments, the layer of insulating material **102** includes silicon oxide (SiOx), doped silicate glass, and/or low-k dielectrics.

[0037] Referring to FIG. 2A, the via **104** and the trench **105** are formed in the layer of insulating material **102**. A plurality of photolithography, deposition, dry etch, and wet etch steps can be employed to define the patterns and form the via **104** and the trench **105** in the layer of insulating material **102**. In some embodiments, the via **104** and the trench **105** are formed using separate etch steps. In some embodiments, the via **104** and the trench **105** are formed using the same etch step.

[0038] Referring to FIG. 2B, the layer of via-plus-trench barrier **106** is formed selectively on the sidewall of the trench **105**, the bottom of the trench **105**, and the sidewall of the via **104**, according to some embodiments. Minimal or no via-plus-trench barrier is formed over the overlapping area between the via **104** and the under-layer conductor **101**. Various exemplary methods of selectively forming the via-plus-trench barrier **106** illustrated by FIG. 2B are to be discussed in the following paragraphs.

[0039] In some embodiments, the selective formation of the via-plus-trench barrier **106** is realized by a selective deposition of the via-plus-trench barrier **106**. In some embodiments, the via-plus-trench barrier **106** is formed by a CVD, a PECVD, an ALD, or a SAM process that deposits the via-plus-trench barrier **106** on the surface of the insulating materials **102**. In some embodiments, the precursors of the CVD or ALD process have a very high selectivity of reactions between insulating materials (e.g., the layer of insulating material **102**) and conducting materials (e.g., under-layer conductor **101**). For example, the precursor can be selected to have greater affinity to dielectric surfaces. In addition, a passivation layer can be absorbed on the metal surfaces to enhance the selectivity of the CVD or ALD process. The passivation layer can be, for example, organic acid, thiol, amine, phosphine, other suitable passivation layer, or combinations thereof. As a result, the deposition of the via-plus-trench barrier **106** occurs on the insulating material **102** but the deposition is minimal on the under-layer conductor **101**. In some embodiments, the via-plus-trench barrier **106** is not deposited on the under-layer conductor **101**. In some embodiments, the processing temperature can be in a range between about 10° C. and about 400° C.

[0040] In some embodiments, the via-plus-trench barrier **106** is a Si containing diffusion barrier, and the precursors of the CVD, PECVD or ALD process contain Si or Si compounds. The CVD, PECVD or ALD process is an oxidizing process that favors Si surface with dangling Si bonds, Si—H bond, Si—C bond, or Si—O bond. The oxidizing process is inhibited on a metallic surface or a highly doped Si surface where the dangling bonds do not exist. The precursor molecules favorably adsorbs on the insulating material **102** with dangling Si bonds, Si—H bond or Si—O bond and

reacts with the insulating material surface to create the layer of via-plus-trench barrier **106**. Meanwhile, minimal to no reaction occurs on the surface of the under-layer conductor **101**, thus minimal to no via-plus-trench barrier **106** is formed on the under-layer conductor **101**. Examples of Si containing diffusion barrier layer can be organosilicate, silicon nitride, silicon carbide, silicon carbon nitride, silicon oxycarbide, other suitable material, and/or combinations thereof.

[0041] In some embodiments, the via-plus-trench barrier **106** is enabled by a selective self-assembled-monolayer (SAM) process. In some embodiments, the selective SAM process is a vapor phase or a liquid phase process that forms one or more monolayers of molecules of SAM material on the surface of the insulating material **102**. In some embodiments, the SAM material includes an organo-silane or an organo-phosphane, and the SAM material acts as the via-plus-trench barrier **106**. In some embodiments, the SAM material acts as a nucleation layer or an enhancer layer to initiate or enhance the subsequent formation of the via-plus-trench barrier **106**. In some embodiments, the selective SAM process forms one or more monolayers of molecules on the surface of the under-layer conductor **101**. The selective SAM material acts as an inhibitor to subsequent barrier deposition process. Due to the existence of the selective SAM material, the via-plus-trench barrier **106** deposition is delayed or inhibited on the surface of conducting material. As a result, the via-plus-trench barrier **106** is formed on the surface of the insulating materials **102**, while minimal or no via-plus-trench barrier **106** is formed on the surface of the under-layer conductor **101**.

[0042] According to some embodiments of the present disclosure, the process temperatures of the CVD, PECVD, ALD or SAM process for forming the via-plus-trench barrier **106** is at or below 400° C. and greater than 10° C.

[0043] After the selective formation of via-plus-trench barrier **106**, the via **104** and the trench **105** are filled with conducting materials. In some embodiments, the via **104** and the trench **105** are filled using separate steps with a single damascene process.

[0044] Referring to FIG. 2C, in a single damascene process, the via **104** is first filled with the via conductor **107**. In some embodiments, the formation of the via conductor **107** in the via **104** is a selective deposition process. Such selective deposition process deposits the via conductor **107** selectively on the bottom of the via **104** where there is an overlapping area between the bottom of the via **104** and the under-layer conductor **101**. As a result, the deposition of the via conductor **107** progresses in a bottom-up fashion that gradually fills the via **104** from the bottom of the via to the top of the via.

[0045] In some embodiments, the selective deposition process includes a CVD process, an ALD process, or an electroless deposition (ELD) process. In some embodiments, the selective deposition process is a selective metal deposition process that fills the via with metal in a bottom-up fashion. For example, copper (Cu) can be selectively deposited in a bottom-up fashion using ELD. In some embodiments, the selective deposition process is a selective CVD process that fills the via with metal in a bottom-up fashion. For example, cobalt (Co) can be formed in a bottom-up fashion using selective CVD. In some embodiments, the selectivity of the selective CVD Co process is realized by choosing an appropriate precursor that forms a nucleation

layer on the surface of the under-layer conductor **101**. In some embodiments, the Co precursor includes a Co containing compound that has —CO— groups and organic ligands in its molecule structure.

[0046] In some embodiments, as illustrated by FIG. 2C, the deposition of the via conductor **107** stops when the top surface of the via conductor **107** is at or about at the same level of the top opening of the via **104**. The top surface of the via conductor **107** can also be at a level that is higher or lower than the top opening of the via **104**. The different levels of the top surface of the via conductor **107** can be due to design or due to process variation.

[0047] In some embodiments, the via conductor **107** includes Cu, Co, nickel (Ni), ruthenium (Ru), rhodium (Rh), iridium (Ir), osmium (Os), aluminum (Al), indium (In), tungsten (W), or carbon nanotubes (CNTs). In some embodiments, a catalytic layer such as Cu is deposited in the via followed by a CVD process with carbon source such as, for example, methane, ethylene, acetylene, other suitable carbon sources, and/or combinations thereof.

[0048] Referring to FIG. 2D, after the formation of the via conductor **107** in the via **104**, the trench barrier **109** is formed on the upper surface of the trench **105**, the sidewall surfaces of the trench **105** and the bottom of the trench **105**. In some embodiments, the trench barrier **109** is in contact with the via conductor **107** at the overlapping area between the trench **105** and the via **104**. In some embodiments, the formation of the trench barrier **109** is enabled by a PVD process, a CVD process, an ALD process, or an ELD process. In some embodiments, the process temperature for forming the trench barrier **109** is at or below 400° C. and greater than 10° C.

[0049] In some embodiments, the trench barrier **109** includes a metal (e.g., Ta, TiW, and/or other metal or metal alloys), a metal oxide (e.g., alumina, manganese oxide, chromium oxide, niobium oxide, titanium oxide, and/or combinations thereof), a metal nitride (e.g. TaN, TiN), a metal compound (e.g., alumina, manganese oxide, chromium oxide, niobium oxide, titanium oxide, and/or combinations thereof), a carbon containing material, other suitable materials, and/or combinations thereof. In some embodiments, the thickness of the trench barrier **109** is between 1 Å and 50 Å. In some embodiments, the thickness of the trench barrier **109** thickness is between 10 Å and 30 Å. In some embodiments, the thickness of the trench barrier **109** thickness is between 15 Å and 25 Å.

[0050] Referring to FIG. 2E, subsequent to the formation of the trench barrier **109**, the trench conductor **108** is formed in the trench **105**, according to some embodiments. In some embodiments, the formation process of the trench conductor **108** includes a PVD process, a CVD process, an ALD process, or an ELD process. In some embodiments, the trench conductor **108** includes Cu, Co, Ni, Ru, Rh, Ir, Os, Al, In, Ag, Au, W, or carbon nanotubes.

[0051] Referring to FIG. 2F, in some embodiments, the excess via-plus-trench barrier **106**, trench barrier **109**, and trench conductor **108** over the upper surface of the insulating material **102** are removed by a CMP process. As a result, the upper surface of the insulating material **102** is planarized and no excessive conducting material is left on the upper surface of the insulating material **102**.

[0052] In some embodiments, an interconnect can be formed in the via and trench using a dual damascene process. FIGS. 3A to 3C are cross sectional views of an

exemplary interconnect structure formed with the dual damascene process, according to some embodiments. The interconnect structure is formed on the substrate **100** (not shown in FIGS. 3A to 3C).

[0053] Referring to FIG. 3A, the layer of via-plus-trench barrier **106** is formed selectively on the sidewall of the trench **105**, the bottom of the trench **105**, and the sidewall of the via **104**, according to some embodiments. Minimal or no via-plus-trench barrier is formed over the overlapping area between the via **104** and the under-layer conductor **101**. Various exemplary methods of selectively forming the via-plus-trench barrier **106** illustrated by FIG. 3A are previously discussed in FIG. 2B.

[0054] Referring to FIG. 3B, in the dual damascene process, the via **104** and trench **105** are filled with a via-plus-trench conductor **301**. In some embodiments, the formation process of the via-plus-trench conductor **301** includes a PVD process, a CVD process, an ALD process, and/or an ELD process. In some embodiments, the via-plus-trench conductor includes Cu, Co, Ni, Ru, Rh, Ir, Os, Al, In, Ag, Au, W, or CNTs.

[0055] Referring to FIG. 3C, in some embodiments, the excess via-plus-trench barrier **106** and via-plus-trench conductor **301** over the upper surface of the insulating material **102** are removed by a CMP process. As a result, the upper surface of the insulating material **102** is planarized and no excessive conducting material is left on the upper surface of the insulating material **102**. Unlike the structure discussed in FIG. 2F where the trench conductor **108** and via conductor **107** are electrically coupled via a trench barrier **106**, the interconnect structure in FIG. 3C has one via-plus-trench conductor **301** for both the trench **105** and the via **104** with no interface between the trench conductor and via conductor. The removal of the interface between the trench conductor and the via conductor can result in a reduction in contact resistance between the trench conductor and the via conductor, according to some embodiments.

[0056] In some embodiments, the selective formation of the via-plus-trench barrier can be formed by a non-selective deposition of the via-plus-trench barrier followed by a selective removal of a portion of the via-plus-trench barrier. FIGS. 4A to 4G are cross sectional views of forming exemplary interconnect structures with a selective removal of a portion of the via-plus-trench barrier.

[0057] Referring to FIG. 4A, the via **104** and the trench **105** are formed in the layer of insulating material **102**. A plurality of photolithography, deposition, dry etch, and wet etch steps are employed to define the patterns and form the via **104** and the trench **105** in the layer of insulating material **102**. In some embodiments, the via **104** and the trench **105** are formed by separate etch steps. In some embodiments, the via **104** and the trench **105** are formed by the same etch step.

[0058] Referring to FIG. 4B, the via-plus-trench barrier **401** is formed non-selectively on the surface of the insulating materials **102** and on the surface of the under-layer conductor **101**. A bottom portion of the barrier **401** (d-d') is in contact with a portion of the top surface of the under-layer conductor **101**. In some embodiments, the formation process of the via-plus-trench barrier **401** includes ALD, CVD, and/or PVD. In some embodiments, the temperature for the formation process of the via-plus-trench barrier **401** is below 400° C. and greater than 10° C.

[0059] Referring to FIG. 4C, the bottom portion (d-d') of the via-plus-trench barrier **401** is selectively removed,

according to some embodiments. In some embodiments, the selective removal can be done with a thermal alloying process. The thermal alloying process can cause an alloy process between the bottom portion (d-d') of the via-plus-trench barrier **401** and the top surface of the under-layer conductor **101**. After the alloying process, the bottom portion (d-d') of the via-plus-trench barrier **401** alloys with the top surface of the under-layer conductor **101**, leaving the no via-plus-trench barrier on the surfaces of the under-layer conductor **101**.

[0060] In some embodiments, the selective removal of a bottom portion (d-d') of the via-plus-trench barrier **401** is done by a lift-off process. Prior to the deposition of a via-plus-trench barrier, a layer of sacrificial material can be selectively deposited on the top surface of the under-layer conductor **101** that is open to the bottom of the via **104**. A via-plus-trench barrier is then non-selectively deposited on the surface of the insulating material **102** and on the top surface of the sacrificial material. The layer of sacrificial material is subsequently etched in an etchant. In some embodiments, the sacrificial material **303** is an organic SAM layer (e.g., thiol derivatives), and the etchant is an organic solvent (e.g., ethanol) that attacks the organic SAM layer but not the via-plus-trench barrier **401**, the under-layer conductor **101**, or the insulating material **102**. The etching of the sacrificial material lifts off the bottom portion (d-d') of the via-plus-trench barrier **401**. As a result, the bottom portion (d-d') of the via-plus-trench barrier **401** is selectively removed.

[0061] Referring to FIG. 4D, in a single damascene process, the via **104** is filled with a via conductor **402**, according to some embodiments. In some embodiments, the formation of the via conductor **402** in the via **104** is a selective deposition process. Such selective deposition process deposits the via conductor **402** selectively on the bottom of the via **104** where there is an overlapping area between the bottom of the via **104** and the under-layer conductor **101**. As a result, the deposition of the via conductor **402** progresses in a bottom-up fashion that gradually fills the via **104** from the bottom of the via to the top of the via.

[0062] In some embodiments, the selective deposition process includes a CVD process, an ALD process, or an electroless deposition (ELD) process. In some embodiments, the selective deposition process is a selective metal deposition process that fills the via with in a bottom-up fashion. In some embodiments, Cu is deposited in a bottom-up fashion. In some embodiments, the selective deposition process is a selective CVD process that fills the via with metal in a bottom-up fashion. In some embodiments, Co is deposited in a bottom-up fashion using the selective CVD process. In some embodiments, the selectivity of the selective CVD Co process is realized by choosing an appropriate precursor that forms a nucleation layer on the surface of the under-layer conductor **101**. In some embodiments, the Co precursor includes a Co containing compound that has —CO— groups and organic ligands in its molecule structure.

[0063] In some embodiments, as illustrated by FIG. 4D, the deposition of the via conductor **402** stops when the top surface of the via conductor **402** is at or about at the same level of the top opening of the via **104**. The top surface of the via conductor **402** can also be at a level that is higher or lower than the top opening of the via **104**. The different levels of the top surface of the via conductor **402** can be due

to design or due to the process variation. In some embodiments, the via conductor 402 includes Cu, Co, Ni, Ru, Rh, Ir, Os, Al, In, W, or CNTs.

[0064] Referring to FIG. 4E, after the formation of the via conductor 402 in the via 104, a trench barrier 403 is formed on the upper surface of the trench 105, the sidewall surfaces of the trench 105, and the bottom of the trench 105. In some embodiments, the trench barrier 403 is in contact with the via conductor 402 at the overlapping area between the trench 105 and the via 104. In some embodiments, the formation of the trench barrier 403 is enabled by a PVD process, a CVD process, an ALD process, or an ELD process. In some embodiments, the process temperature for forming the trench barrier 403 is at or below 400° C. and greater than 10° C.

[0065] In some embodiments, the trench barrier 403 includes a metal (e.g., tantalum (Ta)), a metal compound (e.g., titanium nitride (TiN), tantalum nitride (Ta₃N₅)), and/or a carbon containing material (e.g., graphene). In some embodiments, the thickness of the trench barrier 403 is between 1 Å and 50 Å. In some embodiments, the thickness of the trench barrier 403 thickness is between 10 Å and 30 Å. In some embodiments, the thickness of the trench barrier 403 thickness is between 15 Å and 25 Å. In some embodiments, the thickness of the trench barrier 403 thickness is between 1 Å and 10 Å.

[0066] Referring to FIG. 4F, subsequent to the formation of the trench barrier 402, a trench conductor 404 is formed in the trench 105, according to some embodiments. In some embodiments, the formation process of the trench conductor 404 includes a PVD process, a CVD process, an ALD process, or an ELD process. In some embodiments, the trench conductor 404 includes Cu, Co, Ni, Ru, Rh, Ir, Os, Al, In, W, or carbon nanotubes.

[0067] Referring to FIG. 4G, in some embodiments, the excess via-plus-trench barrier 401, trench barrier 403 and trench conductor 404 over the upper surface of the insulating material 102 are removed by a CMP process. As a result, the upper surface of the insulating material 102 is planarized and no excessive conducting material is left on the upper surface of the insulating material 102.

[0068] In some embodiments, an interconnect can be formed in the via and trench-using a non-selective deposition of the via-plus-trench barrier followed by a selective removal of a portion of the via-plus-trench barrier via a dual damascene process. FIGS. 5A to 5C are cross sectional views of an exemplary interconnect structure formed with a dual damascene process. The interconnect structure is formed on the substrate 100 (not shown in FIGS. 5A to 5C).

[0069] Referring to FIG. 5A, the via-plus-trench barrier 401 is formed by non-selectively depositing the via-plus-trench barrier 401 and selectively removing the bottom portion of the via-plus-trench barrier 401. Various exemplary methods of selectively removing the bottom portion of the via-plus-trench barrier 401 are discussed previously with respect to FIGS. 4B and 4C above.

[0070] Referring to FIG. 5B, in the dual damascene process, the via 104 and trench 105 are filled with a via-plus-trench conductor 501. In some embodiments, the formation process of the via-plus-trench conductor 501 includes a PVD process, a CVD process, an ALD process, and/or an ELD process. In some embodiments, the via-plus-trench conductor 501 includes Cu, Co, Ni, Ru, Rh, Ir, Os, Al, In, W, or CNTs.

[0071] Referring to FIG. 5C, in some embodiments, the excess via-plus-trench barrier 401 and via-plus-trench conductor 501 over the upper surface of the insulating material 102 are removed by a CMP process. As a result, the upper surface of the insulating material 102 is planarized and no excessive conducting material is left on the upper surface of the insulating material 102. Unlike the structure discussed in FIG. 4G where the trench conductor 404 and via conductor 402 are electrically coupled via a trench barrier 403, the interconnect structure in FIG. 5C has one via-plus-trench conductor 501 for both the trench 105 and the via 104 with no interface between the trench conductor and via conductor. The removal of the interface between the trench conductor and the via conductor can result in a reduction in contact resistance between the trench conductor and the via conductor, according to some embodiments.

[0072] FIGS. 6A to 6E are cross sectional views of forming exemplary interconnect structures without forming a selective via-plus-trench barrier, according to some embodiments.

[0073] Referring to FIG. 6A, the via 104 and the trench 105 are formed in the layer of insulating material 102 on the substrate 100 (not shown in FIGS. 6A to 6E). A plurality of photolithography, deposition, dry etch, and wet etch steps can be employed to define the patterns and form the via 104 and the trench 105 in the layer of insulating material 102. In some embodiments, the via 104 and the trench 105 are formed by separate etch steps. In some embodiments, the via 104 and the trench 105 are formed by the same etch step.

[0074] Referring to FIG. 6B, instead of forming a via-plus-trench barrier, the via 104 is directly filled with a via conductor 601, according to some embodiments. In some embodiments, the formation of the via conductor 601 in the via 104 is a selective deposition process. Such selective deposition process deposits the via conductor 601 selectively on the bottom of the via 104 where there is an overlapping area between the bottom of the via 104 and the under-layer conductor 101. As a result, the deposition of the via conductor 601 progresses in a bottom-up fashion that gradually fills the via 104 from the bottom of the via to the top of the via.

[0075] In some embodiments, the selective deposition process includes a CVD process, an ALD process, or an electroless deposition (ELD) process. In some embodiments, the via conductor 601 is a material that does not diffuse into the insulating material 102. In some embodiments, the via conductor 601 includes Ru, Co, and/or CNTs. In some embodiments, the selective deposition process is a selective metal deposition process that fills the via with metal in a bottom-up fashion. In some embodiments, Ru can be deposited in a bottom-up fashion. In some embodiments, the selective deposition process is a selective CVD process that fills the via with metal in a bottom-up fashion. For example, cobalt (Co) can be deposited in a bottom-up fashion using a selective CVD process. In some embodiments, the selectivity of the selective CVD Co process is realized by choosing an appropriate precursor that forms a nucleation layer on the surface of the under-layer conductor 101. In some embodiments, the Co-containing precursor includes a Co containing compound that has —CO— groups and organic ligands in its molecule structure.

[0076] In some embodiments, as illustrated by FIG. 6B, the deposition of the via conductor 601 stops when the top surface of the via conductor 601 is at or about at the same

level of the top opening of the via **104**. The top surface of the via conductor **601** can also be at a level that is higher or lower than the top opening of the via **104**. The different levels of the top surface of the via conductor **601** can be due to design or due to the process variation.

[0077] Referring to FIG. 6C, after the formation of the via conductor **601** in the via **104**, a trench barrier **602** is formed on the upper surface of the trench **105**, the sidewall surfaces of the trench **105** and the bottom of the trench **105**. In some embodiments, the trench barrier **602** is in contact with the via conductor **601** at the overlapping area between the trench **105** and the via **104**. In some embodiments, the formation of the trench barrier **602** is enabled by a PVD process, a CVD process, an ALD process, or an ELD process. In some embodiments, the process temperature for forming the trench barrier **602** is at or below 400° C. and greater than 10° C.

[0078] In some embodiments, the trench barrier **602** includes a metal (e.g., Ta, TiW, and/or other metal or metal alloys), a metal oxide (e.g., alumina, manganese oxide, chromium oxide, niobium oxide, titanium oxide, and/or combinations thereof), a metal nitride (e.g. TaN, TiN), a metal compound (e.g., alumina, manganese oxide, chromium oxide, niobium oxide, titanium oxide, and/or combinations thereof), a carbon containing material, other suitable materials, and/or combinations thereof. In some embodiments, the thickness of the trench barrier **602** is between 1 Å and 50 Å. In some embodiments, the thickness of the trench barrier **602** thickness is between 10 Å and 30 Å. In some embodiments, the thickness of the trench barrier **602** thickness is between 15 Å and 25 Å. In some embodiments, the thickness of the trench barrier **602** thickness is between 1 Å and 10 Å.

[0079] Referring to FIG. 6D, subsequent to the formation of the trench barrier **602**, a trench conductor **603** is formed in the trench **105**, according to some embodiments. In some embodiments, the formation process of the trench conductor **603** includes a PVD process, a CVD process, an ALD process, or an ELD process. In some embodiments, the trench conductor **603** includes Cu, Co, Ni, Ru, Rh, Ir, Os, Al, In, W, or carbon nanotubes.

[0080] Referring to FIG. 6E, in some embodiments, the excess trench barrier **602** and trench conductor **603** over the upper surface of the insulating material **102** are removed by a CMP process. As a result, the upper surface of the insulating material **102** is planarized and no excessive conducting material is left on the upper surface of the insulating material **102**.

[0081] FIGS. 7A to 7D are cross sectional views of forming exemplary interconnect structures with a self-forming barrier via a dual damascene process, according to some embodiments.

[0082] Referring to FIG. 7A, the via **104** and the trench **105** are formed in the layer of insulating material **102** on the substrate **100** (not shown in FIGS. 7A to 7D). A plurality of photolithography, deposition, dry etch, and wet etch steps are employed to define the patterns and form the via **104** and the trench **105** in the layer of insulating material **102**. In some embodiments, the via **104** and the trench **105** are formed by separate etch steps. In some embodiments, the via **104** and the trench **105** are formed by the same etch step.

[0083] Referring to FIG. 7B, in some embodiments, the via **104** and the trench **105** are filled at the same time with the dual damascene process. In the dual damascene process,

the via **104** and trench **105** are filled with a via-plus-trench conductor **701**. The via-plus-trench conductor **701** is in contact with the insulating material **102**. In some embodiments, filling the via **104** and trench **105** with the via-plus-trench conductor **701** includes a PVD process, a CVD process, an ALD process, and/or an ELD process. In some embodiments, the via-plus-trench conductor **701** includes a metal (e.g., Cu) alloyed with B, Al, Ti, Cr, Mn, Nb, Pd, tin (Sn), zinc (Zn), magnesium (Mg), Ag, In, or Au.

[0084] Referring to FIG. 7C, in some embodiments, the via-plus-trench conductor **701** is planarized with a CMP process, leaving no excess via-plus-trench conductor **701** on the upper surface of the insulating material **102**.

[0085] Referring to FIG. 7D, subsequent to the planarization of the via-plus-trench conductor **701**, a self-forming barrier **702** is formed on the sidewall surfaces of the trench **105**, the bottom of the trench **105**, and the sidewall surfaces of the via **104**. In some embodiments, the self-forming barrier **702** is formed by treating the interconnect structure with a thermal or an electrical treatment process. In some embodiments, the treatment process causes the via-plus-trench conductor **701** or the dopants in the via-plus-trench conductor **701** to react with the insulating material **102**. In some embodiments, the treatment drives the dopants via thermal diffusion or electrical drifting to the interface between the insulating material **102** and the via-plus-trench conductor **701**, and the dopants subsequently form a diffusion barrier.

[0086] In some embodiments, the via-plus-trench conductor **701** can be Cu alloyed with Mn, where the Mn has a weight percentage in the alloy between 0.5% and 10%. The under-layer conductor **101** can be Cu. In some embodiments, the insulating material **102** is silicon dioxide (SiO₂). With a thermal treatment between 250° C. and 400° C., Mn diffuses to the interface between the via-plus-trench conductor **701** and insulating materials **102**, and Mn further reacts with SiO₂ in the insulating material **102** and forms a layer of self-forming barrier **702**, as illustrated by FIG. 7D. In some embodiments, the self-forming barrier **702** is in a form of manganese oxide (MnOx) or manganese silicon oxide (MnSiOx). The self-forming barrier **702** is not formed at the bottom of the via **104** because Mn does not form MnOx or MnSiOx with the under-layer conductor **101** (e.g., Cu). As a result, the self-forming barrier **702** is formed on the sidewall of the trench **105**, the bottom of the trench **105**, and the sidewall of the via **104**. No self-forming barrier **702** is formed on the upper surface of the under-layer conductor **101** at the bottom of the via **104**.

[0087] FIG. 8 is a flow diagram of an example method **800** of forming exemplary interconnect structures, according to some embodiments. Based on the disclosure herein, other operations in method **800** can be performed. Further, the operations of method **800** can be performed in a different order and/or vary.

[0088] At operation **801**, an under-layer conductor is formed on and/or within a semiconductor substrate. The semiconductor structure can include bulk silicon (e.g., doped or undoped silicon) or an active layer of a silicon-on-insulator (SOI) substrate. The semiconductor structure can include semiconductor material such as, for example, silicon, germanium, silicon germanium, silicon germanium on insulator (SGOI), or combinations thereof. Active devices can be formed on and/or within the semiconductor

substrate. Active and passive devices such as, for example, transistors, diodes, capacitors, resistors, inductors, and the like can be formed.

[0089] In some embodiments, the under-layer conductor includes copper (Cu), cobalt (Co), nickel (Ni), ruthenium (Ru), rhodium (Rh), iridium (Ir), osmium (Os), aluminum (Al), indium (In), silver (Ag), gold (Au), tungsten (W), or carbon nanotubes. In some embodiments, the under-layer conductor includes metals alloyed with boron (B), aluminum (Al), titanium (Ti), chromium (Cr), manganese (Mn), niobium (Nb), palladium (Pd), silver (Ag), indium (In), tin (Sn), zinc (Zn), or gold (Au).

[0090] At operation **802**, a layer of insulating material is formed over the under-layer conductor. In some embodiments, the layer of insulating material is formed by chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), a spin-on process, physical vapor deposition (PVD), or Atomic Layer Deposition (ALD). In some embodiments, the layer of insulating material includes silicon oxide (SiO_x), doped silicate glass, and/or various low-k dielectrics.

[0091] At operation **803**, a via and a trench is formed within the insulating material. A plurality of photolithography, deposition, dry etch, and wet etch steps can be employed to define the patterns and form the via and the trench in the layer of insulating material. In some embodiments, the via and the trench are formed by separate etch steps. In some embodiments, the via and the trench are formed by the same etch step.

[0092] At operation **804**, the via is filled with a via conductor. According to some embodiments, a single damascene process can be employed to fill the via and the trench in separate steps. In some embodiments, the via is filled with the via conductor in a selective deposition process. Such selective deposition process deposits the via conductor selectively on the bottom of the via where there is an overlapping area between the bottom of the via and the under-layer conductor. As a result, the deposition of the via conductor inside the via progresses in a bottom-up fashion that gradually fills the via from the bottom of the via to the top of the via. According to some embodiments, due to the selective deposition of the via-conductor, there is no barrier between the via conductor and the under-layer conductor, and the via conductor is in contact with the under-layer conductor.

[0093] In some embodiments, the selective deposition of the via conductor includes a CVD process, an ALD process, or an electroless deposition (ELD) process. In some embodiments, the selective deposition of the via conductor is a selective metal deposition process that fills the via with metal in a bottom-up fashion. In some embodiments, Cu is deposited in a bottom-up fashion. In some embodiments, the selective deposition of the via conductor is a selective CVD metal process that fills the via with metal in a bottom-up fashion. In some embodiments, Co is deposited in a bottom-up fashion using the selective CVD metal process. In some embodiments, the selectivity of the selective CVD Co process is realized by choosing an appropriate precursor that forms a nucleation layer on the surface of the under-layer conductor **101**. In some embodiments, the Co precursor includes a Co containing compound that has —CO— groups and organic ligands in its molecule structure.

[0094] At operation **805**, the trench is filled with a trench conductor. In some embodiments, the formation process of

the trench conductor includes a PVD process, a CVD process, an ALD process, or an ELD process. In some embodiments, the trench conductor includes Cu, Co, Ni, Ru, Rh, Ir, Os, Al, In, W, or carbon nanotubes.

[0095] In some embodiments, prior to filling the trench with the trench conductor, a trench barrier can be formed on the upper surface of the trench, the sidewall surfaces of the trench and the bottom of the trench. In some embodiments, the trench barrier is in contact with the via conductor at the overlapping area between the trench and the via. In some embodiments, the formation of the trench barrier is performed by a PVD process, a CVD process, an ALD process, or an ELD process. In some embodiments, the process temperature for forming the trench barrier is at or below 400° C. and greater than about 10° C.

[0096] The absence of a barrier between the under-layer conductor and the via conductor provides benefits to the interconnect structures. These benefits include a reduction in contact resistance between the via conductor and the under-layer conductor, thereby reducing the overall interconnect structure resistance and improving the speed of the device. In addition, in some embodiments, the absence of a barrier at the bottom and the sidewall of the via provides additional room for the via conductor and reduces the resistance of the via conductor, which also reduces the overall resistance of the interconnect structure.

[0097] In some embodiments, an interconnect structure includes a first layer of conducting material. The interconnect structure can also include a layer of insulating material above the first layer of conducting material. The interconnect structure also includes a via formed in the layer of insulating material. A bottom of the via is open to the first layer of conducting material. The interconnect structure also includes a trench formed above the via in the layer of insulating material. The interconnect structure also includes a second layer of conducting material formed in the via. The second layer of conducting material is in contact with the first layer of conducting material. The interconnect structure also includes a third layer of conducting material formed in the trench.

[0098] In some embodiments, a method of fabricating an interconnect structure includes forming a first layer of conducting material. A layer of insulating material is formed over the first layer of conducting material. A via is formed within the insulating material, and a bottom of the via is open to the first layer of conducting material. A trench is formed above the via and in the insulating material. The via is filled with a second layer of conducting material, and the second layer of conducting material is in contact with the first layer of conducting material. The trench is filled with a third layer of conducting material.

[0099] In some embodiments, an interconnect structure includes a first layer of conducting material. The interconnect structure can also include a layer of insulating material above the first layer of conducting material. The interconnect structure also includes a via formed in the layer of insulating material. A bottom of the via is open to the first layer of conducting material. The interconnect structure also includes a trench formed in the layer of insulating material. A portion of the bottom of the trench is in contact with a top of the via. The interconnect structure also includes a second layer of conducting material formed in the via. The second layer of conducting material is in contact with the first layer of conducting material. The interconnect structure

also includes a third layer of conducting material formed in the trench. A portion of a bottom of the third layer of conducting material overlaps with the top of the via. The interconnect structure also includes a via-plus-trench barrier formed on a sidewall of the via, a sidewall of the trench, and a bottom of the trench.

[0100] It is to be appreciated that the Detailed Description section, and not the Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections may set forth one or more but not all exemplary embodiments of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the present disclosure and the appended claims in any way.

[0101] The foregoing disclosure outlines features of several embodiments so that a person of ordinary skill in the art may better understand the aspects of the present disclosure. A person of ordinary skill in the art will appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. A person of ordinary skill in the art will also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

1.-20. (canceled)

21. A structure, comprising:

- a conductive layer;
- a dielectric structure on the conductive layer;
- a conductive via in the dielectric structure, wherein the conductive via is barrier free, and wherein a top surface of the conductive via is wider than a bottom surface of the conductive via;
- a first conductive structure comprising a barrier layer, disposed above the conductive via and within the dielectric structure, wherein the barrier layer is disposed on a top surface of the conductive via; and
- a second conductive structure within the dielectric structure, wherein the second conductive structure is laterally displaced from the conductive via.

22. The structure of claim 21, wherein the conductive via is in contact with the conductive layer.

23. The structure of claim 21, wherein a bottom surface of the first conductive structure is substantially coplanar with a bottom surface of the second conductive structure.

24. The structure of claim 21, wherein the first conductive structure comprises a metal layer.

25. The structure of claim 24, wherein the barrier layer is disposed between the metal layer and the dielectric structure.

26. The structure of claim 21, wherein the conductive via is in contact with the barrier layer.

27. The structure of claim 21, wherein the conductive layer extends below the first and the second conductive structures.

28. A structure, comprising:

- a layer of conductive material formed on source/drain areas of a field-effect-transistor (FET) device;
- an insulating layer interfacing with the layer of conductive material;

- a via in the insulating layer and formed on the layer of conductive material, wherein the via comprises a conductive layer interfacing with the insulating layer, and wherein the conductive layer comprises an elemental metal;

- a first conductive structure connected to the via and interfacing with the insulating layer; and

- a second conductive structure laterally displaced from the first conductive structure, wherein the second conductive structure is spaced apart from the layer of conductive material.

29. The structure of claim 28, wherein a bottom surface of the second conductive structure is substantially coplanar with a bottom surface of the first conductive structure.

30. The structure of claim 28, wherein the via has a tapered shape.

31. The structure of claim 30, wherein the via comprises another conductive layer in contact with the conductive layer.

32. The structure of claim 28, wherein a ratio of a height of the via and a height of the first conductive structure is between about 1 and about 20, and wherein the height of the first conductive structure is measured from a top surface of the via.

33. The structure of claim 28, wherein the first conductive structure comprises a first barrier layer, a second barrier layer, and a third conductive layer.

34. The structure of claim 33, wherein the second conductive structure comprises the first barrier layer, the second barrier layer, and the third conductive layer.

35. The structure of claim 33, wherein the second barrier layer is in contact with the third conductive layer.

36. The structure of claim 33, wherein the second barrier layer is disposed between the first barrier layer and the third conductive layer.

37. A method, comprising:

- depositing a layer of insulating material on a layer of a conductive material;

- etching a tapered opening in a first portion of the layer of insulating material;

- depositing a first conductive layer within the tapered opening, wherein the first conductive layer is in contact with the insulating layer;

- forming a trench opening in a second portion of the insulating layer, wherein the second portion is above the first portion; and

- depositing a second conductive layer within the trench opening.

38. The method of claim 37, further comprising substantially co-planarizing a top surface of the second conductive layer with a top surface of the insulating layer.

39. The method of claim 37, further comprising forming a barrier layer within the trench opening prior to depositing the second conductive layer.

40. The method of claim 37, wherein etching the tapered opening in the first portion of the layer of insulating material comprises etching the insulating layer to expose a top surface of the layer of the conductive material.

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